Spectral World Renderer: Unified 3D Hyperspectral Unmixing

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https://ming053l.github.io/Spectral-World-Renderer/

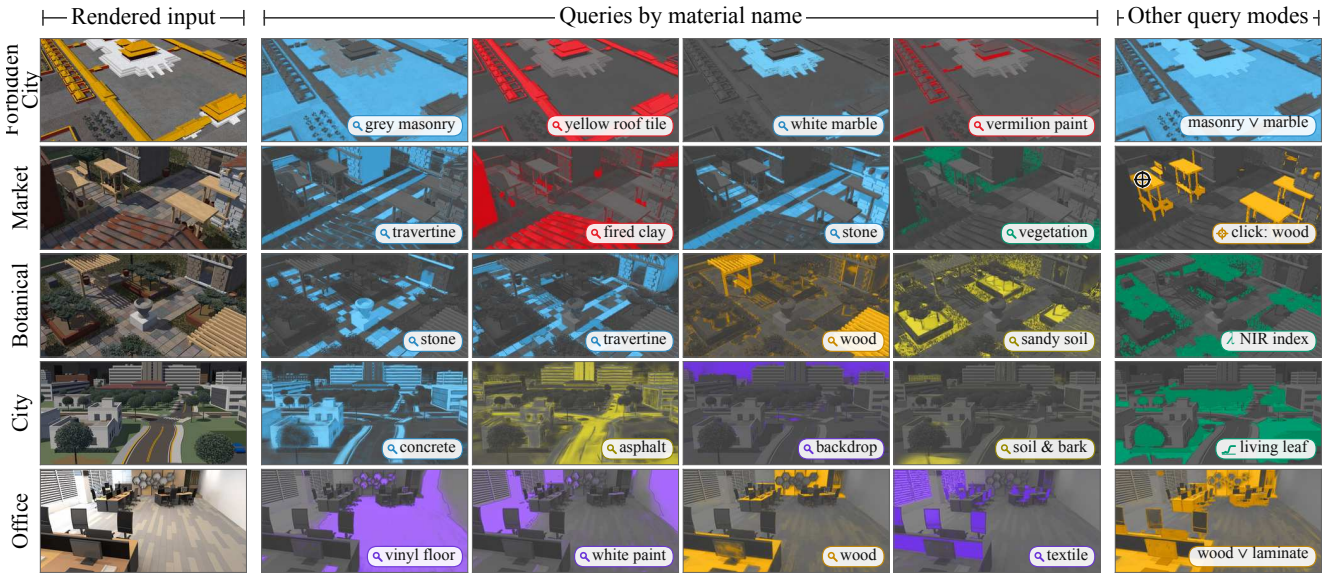


Figure 1. **Material queries across SWR-Bench worlds.** *Input:* multi-view hyperspectral images from generated worlds with known composition (left). *Representation:* Gaussians with material abundances and spectra. *Queries:* a material name mapped from labelled training views, a clicked pixel (crosshair), a reference spectrum, a band index or a combination. Tiles show selections on test views.

Abstract

Surfaces with similar RGB colours can have distinct material spectra. Multi-view hyperspectral imaging offers the evidence to distinguish them and associate their materials with 3D surfaces. Existing 3D unmixing approaches, however, either decompose rendered images after compositing or rely on slow implicit fields. A faithful reconstruction alone does not establish whether its components correspond to materials; this requires multi-view data of known composition. We present Spectral World Renderer (SWR), a framework for world generation and 3D unmixing. A coding agent builds executable worlds with measured material spectra; shared camera samples yield aligned hyperspectral images, material labels and sub-pixel proportions. The resulting SWR-Bench contains eight indoor and outdoor worlds. SWR decodes compact Gaussian features into material abundances before compositing, jointly rendering spectra and material maps. Multi-view appearance regularization discourages distinct appearances from collapsing onto shared endmembers. Fitting uses no material labels; validation labels select the endmember count. SWR achieves the highest mIoU in all eight worlds and the highest mean PSNR on each real dataset evaluated. On City, it trains 3.7–12.9× faster than the unmixing baselines; its optimized renderer averages 34.9 labelled views/s. The fitted primitives support cross-view spectral queries that distinguish surfaces with similar RGB colours.

1. Introduction

Surfaces with similar RGB colours can differ in material spectra, as with green paint and foliage. Distinguishing them matters for robotic sorting, agriculture and inspection. Foundation models recognize objects from RGB and support open-vocabulary 3D queries [39, 42, 61, 68], but object identity does not establish material composition. RGB material labels [4, 50, 85] likewise omit sub-pixel mixing.

Hyperspectral imaging instead records many bands, preserving evidence about material reflectance [20, 22, 43, 54]. Spectral unmixing then models these observations as non-negative mixtures of endmembers [5].

In 3D, however, three challenges arise: spectral dimensionality increases storage, compositing obscures local mixtures, and illumination complicates material assignments.

Metamerism: Spectra Beyond RGB

Different reflectance spectra can yield identical tristimulus values for a fixed illuminant and observer: metamerism [35]. HSI can retain spectral distinctions lost through RGB integration; multi-view learning attaches them to shared 3D primitives for material queries across viewpoints.

Our goal is therefore to reuse spectral distinctions across viewpoints. Shared 3D primitives associate observations of the same surface, allowing a material selection to persist across cameras beyond independent image labels.

Existing spectral reconstruction methods, however, usually leave unmixing to a separate step. Spectral radiance fields and Gaussian splatting [9, 24, 48, 56, 59, 77, 83] represent spectra through per-band quantities or compact latents and primarily evaluate reconstruction fidelity. UnMix-NeRF [64] integrates unmixing into a radiance field, but its implicit representation is costly to train and render and lacks explicit primitives for material selection. Table 1 highlights a further obstacle: evaluation data. Real multi-view hyperspectral datasets [9, 41, 45] provide radiance without material labels or sub-pixel proportions, which are difficult to annotate reliably. Synthetic RGB datasets supply geometry and semantics but lack spectra for evaluating unmixing.

We therefore introduce Spectral World Renderer (SWR), combining world generation with explicit 3D unmixing. As shown in Figure 2, its world generator uses a coding agent to construct executable scenes from reference images and text. Measured spectra define material appearances, while shared camera samples produce aligned RGB and hyperspectral observations, material labels and sub-pixel proportions. The resulting SWR-Bench provides eight worlds in which spectral reconstruction and material identification can be assessed independently: a method may succeed at one while failing at the other.

Within this framework, SWR represents scenes with RGB-initialized Gaussians carrying features of fixed size, independent of band count. At each ray intersection, a shared decoder predicts abundances over a scene-wide endmember dictionary and a separate intensity. Decoding before compositing preserves local mixtures and jointly renders spectra and material maps. Multi-view appearance regularization discourages distinct appearances from collapsing onto the same endmember. Parameter fitting uses no material labels; validation labels select dictionary size. Once fitted, the primitives support cross-view queries by click, spectrum, band or names mapped from training labels.

In brief, our main contributions include executable spectral worlds with composition ground truth; compact Gaussians for pre-compositing unmixing; and multi-view appearance regularization for material allocation. Cross-view queries test the recovered material structure; SWR leads in mIoU across eight worlds.

Table 1. **Ground-truth coverage of multi-view datasets.** ID: material labels; Prop.: geometric area proportions; Gen.: on-demand world generation. NeSpoF-Seg.: synthetic labels from UnMix-NeRF [64]. SWR-Bench pairs images with shared-sample proportions and spectral variability. –: unspecified or variable. *We use the five aligned scenes listed in Table S7.

Dataset	Bands	nm	Scenes ×views	Capture	ID	Prop.	Gen.
Replica	3	RGB	18×–	indoor	✗	✗	✗
Hypersim	3	RGB	461×168	indoor	✗	✗	✗
Kubric	3	RGB	–	random	✗	✗	✓
NeSpoF	21	450–650	4×18–30	tabletop	✗	✗	✗
NeSpoF-Seg.	21	450–650	4×–	synthetic	✓	✗	✗
HS-NeRF	128/141	371–1087	8×48–433*	turntable	✗	✗	✗
HSI-SC	204	397–1004	3×60	turntable	✗	✗	✗
SWR-Bench	64	420–980	8×96–240	in/outdoor	✓	✓	✓

2. Related Work

Hyperspectral 3D reconstruction. Radiance fields use neural or grid models [2, 57, 58]; 3DGS composites Gaussians [38], later improved in quality and speed [25, 32, 53, 72, 92]. In parallel, foundation-model features [42, 68] answer object queries [8, 39, 67, 90, 96] and select materials in RGB [17], and neural harmonic textures vary appearance within a primitive [12]. Hyperspectral fields render radiance per wavelength [9, 41, 45, 48]; Gaussian versions store per-band quantities or compact latents decoded to spectra [24, 56, 59, 77, 83]. Complementary RGB inverse-rendering methods recover reflectance and illumination [6, 51, 94] or reflection components [86]. For material selection, SWR decodes spatially varying features into per-intersection abundances and intensity.

Spectral unmixing. Unmixing explains each spectrum by endmembers and non-negative abundances [5, 26, 40], found by geometric extraction [60, 88], factorization or autoencoders [47, 63, 80], with spatial regularization [36] or spectral-variability and nonlinear models [27, 28]. These 2D abundance maps serve remote sensing and material analysis [20, 66]. Extending this decomposition to 3D, UnMix-NeRF integrates unmixing into a radiance field [64]. Our method likewise learns endmembers and abundances jointly, but uses explicit Gaussians and spatially varying features for material selection.

Programmatic world generation. Procedural rules [69, 70], language-model layouts [16, 89] and scene code [31, 49] build 3D worlds. Programmable and generative world renderers turn scene state or G-buffers into video [33, 34, 52]. To evaluate unmixing, SWR jointly renders measured spectra and material proportions from executable worlds.

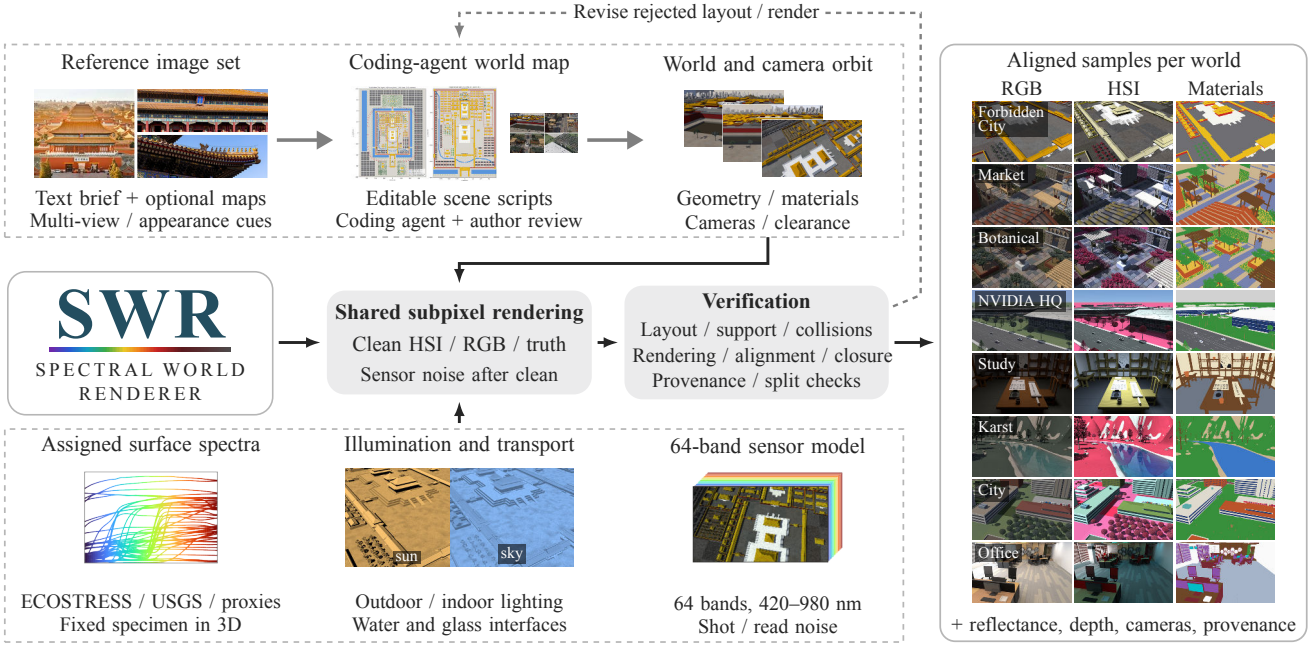


Figure 2. **Reference-guided world generation.** A coding agent converts reference photographs and a text brief into executable geometry, materials and cameras. Measured library spectra define the material appearances. Shared sub-pixel samples yield aligned RGB and hyperspectral observations, material labels and proportions. Layout and rendering checks guide revisions across eight worlds (right).

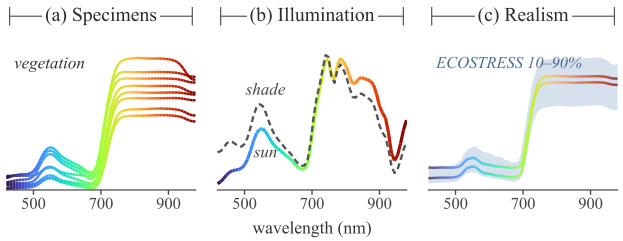


Figure 3. **Material spectra vary with specimen and illumination.** (a) Measured grass specimens have different spectra. (b) Sunlight and shade change the spectral shape of the same grass. (c) Both broadleaf spectra lie within independent leaf measurements.

Synthetic benchmarks and spectral simulation. Synthetic and scanned datasets supply geometry, semantics or intrinsic images [23, 74, 79]; game and agent benchmarks add G-buffers and checkable scene code [19, 34, 75, 91]. Spectral renderers and remote-sensing simulators trace light per wavelength [21, 37, 65, 87], and ray-traced vegetation scenes give known abundances for 2D unmixing [13]. As Table 1 summarizes, these datasets lack paired multi-view spectra and area proportions; SWR-Bench samples both from the same sub-pixel points, varies specimens and illumination, and releases transport coefficients for reproducible target generation.

3. Spectral World Renderer

Spectral World Renderer generates composition targets to evaluate unmixing (Sec. 3.1), decodes material mixtures on Gaussians (Sec. 3.2) and guides their assignments with appearance clusters (Sec. 3.3). Fitting uses no material labels; validation labels select dictionary size K.

A scene is observed from V training cameras. Camera v provides RGB $\mathbf{I}_v \in [0,1]^{H\times W\times 3}$ and a co-registered hyperspectral cube $\mathbf{H}_v \in \mathbb{R}^{H\times W\times L}$ with L bands at resolution $H \times W$. The ray $\mathbf{r} = \mathbf{r}_v(p)$ of pixel p indexes its observed values $\mathbf{I}(\mathbf{r}) \in \mathbb{R}^3$ and $\mathbf{H}(\mathbf{r}) \in \mathbb{R}^L$. Novel views yield spectra $\hat{\mathbf{H}}(\mathbf{r})$ and endmember abundances $\hat{\mathbf{A}}(\mathbf{r}) \in \mathbb{R}^K$ over a shared dictionary of K spectra.

3.1. World Generator and SWR-Bench

Multi-view HSI requires costly spectral calibration and camera registration [9, 41], limiting dataset scale and scene diversity for 3D unmixing. Even a well-registered capture records radiance rather than sub-pixel composition. Additional viewpoints constrain scene structure, yet do not directly supply the material proportions needed to verify unmixing. We therefore generate images and composition targets jointly, as Figure 2 shows.

Reference-guided world generation. To control scene content and acquisition, a coding agent turns reference images, maps and text into programs with editable geometry, materials, lighting and cameras. Coding agents automatically inspect previews for structural support, collisions and clearance, followed by careful human review.

Shared-sample ground truth. To avoid disagreements at silhouettes and mixed pixels, images and labels share the same S sub-pixel camera samples ($S = 16$): for the pixel indexed by ray $\mathbf{r}$, $\mathbf{x}_n$ is the first surface hit of sample n

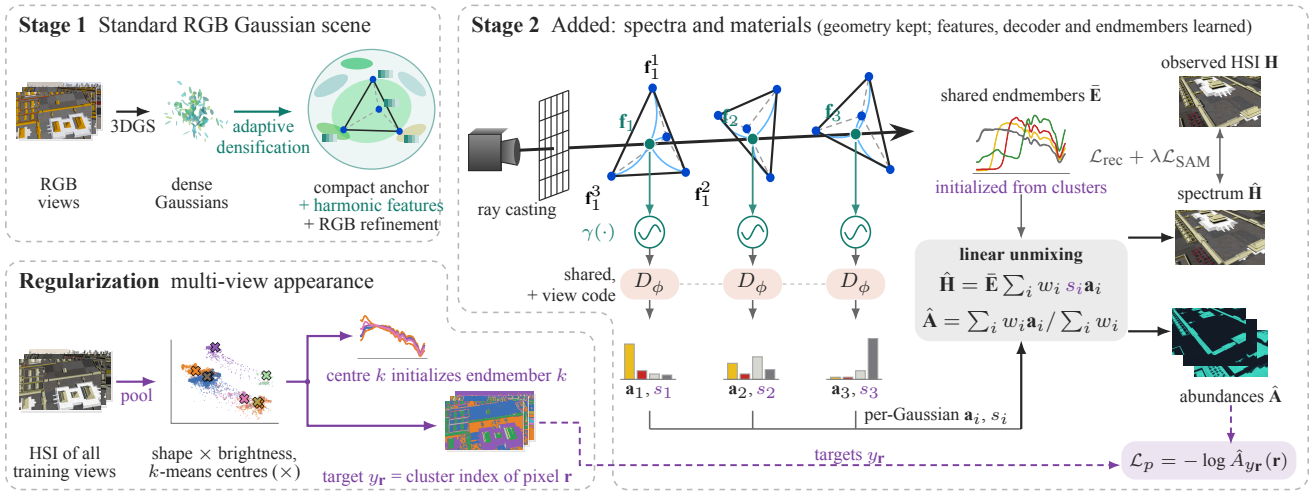


Figure 4. **SWR reconstruction and unmixing pipeline.** Stage 1 fits RGB geometry and harmonic features; Stage 2 freezes geometry and learns spectral mixtures. Per-Gaussian features have fixed size across band counts. At each intersection, a shared decoder predicts abundances and intensity over scene-wide endmembers. Shared visibility weights jointly render spectra and abundances. Appearance clusters initialize the dictionary and regularize abundance assignments.

and $\mathcal{M}(\mathbf{x}_n)$ its material class. A Lambertian specimen j with measured reflectance $\rho_j(\lambda)$ at wavelength λ has, under direct or sky illumination u, band radiance

$$v^u_{j,\ell} = \int \psi_\ell(\lambda)\, \rho_j(\lambda) \Phi_u(\lambda)/\pi \, \mathrm{d}\lambda, \quad u \in \{\mathrm{d}, \mathrm{s}\}, \tag{1}$$

where ψ_ℓ is the response of band ℓ and $\Phi_\mathrm{d}, \Phi_\mathrm{s}$ are direct and sky irradiance. The vectors $\mathbf{v}^u_j \in \mathbb{R}^L$ and shading coefficients $C^u_j(\mathbf{r})$, averaged over the S samples, give

$$\begin{aligned} \mathbf{H}_{\text{clean}}(\mathbf{r}) &= \textstyle\sum_j (C^\mathrm{d}_j \mathbf{v}^\mathrm{d}_j + C^\mathrm{s}_j \mathbf{v}^\mathrm{s}_j), \\ A_c(\mathbf{r}) &= \tfrac{1}{S} \textstyle\sum_{n=1}^{S} [\mathcal{M}(\mathbf{x}_n) = c]. \end{aligned} \tag{2}$$

Here $[\cdot]$ is the indicator function, c indexes C material classes and j measured specimens. Released area proportions renormalize A_c over non-sky samples. Learned endmembers use index k and need not correspond one-to-one to classes. Beyond this formulation, the worlds include richer transport: four outdoor worlds follow this two-term sun/sky Lambertian model; NVIDIA HQ adds rough specular highlights, a glass reflection and a hazy sky; Karst water adds reflection, refraction and absorption; the Study and Office are path traced in Mitsuba [37]. Transport contributions are stored separately for verification.

Measured spectra and variability. ECOSTRESS and USGS provide measured reflectances [43, 54]; recorded assignments preserve spectral provenance. Spectral shape varies with specimens and lighting, as Figure 3(a,b) shows; each surface keeps its specimen across views. Outdoor worlds use ASTM G173 sunlight and skylight [1], interiors white sources [11]. We release clean 64-band cubes at 420–980 nm and versions with shot and read noise [18].

Verification. Finally, we verify label alignment, spectral provenance and split separation. Specifically, stored contributions re-compose clean outdoor radiance, while indoor transport is checked before noise. Figure 3(c) compares assigned spectra with independent library measurements.

3.2. Pre-Compositing Unmixing

To support queries on 3D primitives, we decode mixtures at Gaussian intersections before compositing. Figure 4 shows the joint rendering of spectra and abundances.

RGB anchor and harmonic features. We first establish geometry from RGB, then learn spectra on this shared scaffold. Co-registered RGB views are supplied with the data or derived from visible bands. FastGS densification and pruning [72] retain primitives supported across views. The anchor has N Gaussians G_i with mean $\boldsymbol{\mu}_i$, rotation $\mathbf{R}_i$, scale $\boldsymbol{\sigma}_i$ and opacity o_i; after fitting, we drop colours and fold the rasterizer's screen-space variance into the scales. To retain local detail, each Gaussian carries a neural harmonic texture (NHT) [12] with four feature vectors $\mathbf{f}_{im} \in \mathbb{R}^d$, $m = 1, \ldots, 4$, with $d = 12$. A ray $\mathbf{r}$ queries G_i at the point of maximal Gaussian response along the ray, whose tetrahedral barycentric coordinates $\beta_{im}(\mathbf{r})$ interpolate the vertex features:

$$\mathbf{f}_i(\mathbf{r}) = \sum_{m=1}^{4} \beta_{im}(\mathbf{r}) \mathbf{f}_{im} \in \mathbb{R}^d. \tag{3}$$

Consequently, rays through one Gaussian can decode different mixtures across material boundaries. Before spectral fitting, we refine features and geometry with the NHT RGB shader, without further densification, and then freeze the geometry. Primitive storage is band-independent: 11 geometry/opacity values and 48 features. The shared dictionary adds LK values.

Pre-compositing material decoder. The interpolated feature from Eq. 3, encoded as $\gamma(\mathbf{f}) = [\sin \mathbf{f}; \cos \mathbf{f}] \in \mathbb{R}^{2d}$ [81], enters a shared MLP D_ϕ with a directional code

$\mathbf{Y}(\mathbf{d}_i(\mathbf{r})) \in \mathbb{R}^9$: real spherical harmonics of degree ≤ 2 [71], where $\mathbf{d}_i$ is the unit Gaussian-to-camera direction.

$$\mathbf{z}_i(\mathbf{r}) = D_\phi(\gamma(\mathbf{f}_i(\mathbf{r})), \mathbf{Y}(\mathbf{d}_i(\mathbf{r}))) \in \mathbb{R}^{K+1}. \quad (4)$$

Superscripts index decoder outputs: the first K entries yield abundances $\mathbf{a}_i(\mathbf{r}) = \mathrm{softmax}(\mathbf{z}_i^{1:K}(\mathbf{r}))$; the last gives intensity $s_i(\mathbf{r}) = \mathrm{softplus}(z_i^{K+1}(\mathbf{r}))$. The MLP has three width-128 SiLU hidden layers; a zero-initialized linear map adds the directional code to layer one.

Across intersections, these mixtures share a scene-wide dictionary $\bar{\mathbf{E}} = [\bar{\mathbf{e}}_1, \ldots, \bar{\mathbf{e}}_K] \in \mathbb{R}^{L\times K}$. The parameters $\boldsymbol{\eta}_k \in \mathbb{R}^L$ define each endmember through

$$\bar{\mathbf{e}}_k = \frac{\mathrm{sigmoid}(\boldsymbol{\eta}_k)}{\|\mathrm{sigmoid}(\boldsymbol{\eta}_k)\|_2} \in \mathbb{R}^L_{\geq 0}. \quad (5)$$

With endmember scale fixed by normalization, s_i scales the resulting radiance mixture [5]:

$$\mathbf{h}_i(\mathbf{r}) = s_i(\mathbf{r})\bar{\mathbf{E}}\mathbf{a}_i(\mathbf{r}) \in \mathbb{R}^L_{\geq 0}. \quad (6)$$

Joint spectral and abundance rendering. We then composite both outputs with shared visibility weights. For Gaussians with ray-dependent opacity $\alpha_i(\mathbf{r})$ (from o_i and the Gaussian response), transmittance $T_i(\mathbf{r}) = \prod_{i'<i}(1 - \alpha_{i'}(\mathbf{r}))$ gives visibility weights $w_i(\mathbf{r}) = T_i(\mathbf{r})\alpha_i(\mathbf{r})$ [38]:

$$\hat{\mathbf{H}}(\mathbf{r}) = \sum_i w_i(\mathbf{r})\mathbf{h}_i(\mathbf{r}), \quad \hat{\mathbf{A}}(\mathbf{r}) = \frac{\sum_i w_i(\mathbf{r})\mathbf{a}_i(\mathbf{r})}{\sum_i w_i(\mathbf{r})}. \quad (7)$$

Both $\mathbf{a}_i$ and $\hat{\mathbf{A}}$ are non-negative and sum to one on non-empty rays [26]. However, $\hat{\mathbf{H}} = \bar{\mathbf{E}}\sum_i w_i s_i \mathbf{a}_i$, while $\hat{\mathbf{A}}$ uses visibility weights alone, excluding s_i. A brighter surface can therefore contribute more radiance without receiving more abundance solely from brightness. Nevertheless, these proportions need not equal geometric area fractions. Averaging per-intersection abundances over training rays enables the primitive queries in Sec. 4.4.

3.3. Multi-View Appearance Regularization

Assignment ambiguity. Although Eq. 6 constrains spectral reconstruction, endmember assignments remain ambiguous. Similar materials may share one endmember while s absorbs their brightness difference, as in extended mixing models [14, 28], merging their abundances despite accurate spectra. We therefore guide endmember assignments using the appearance differences in Figure 5.

Dictionary size and material classes. Guidance must also preserve variation across specimens and lighting, as Figure 3 shows. We therefore allow $K > C$ so several endmembers can represent one class [3, 73, 78]. Yet capacity alone can leave distinct materials sharing an endmember, so appearance clusters guide allocation.

Multi-view appearance clusters. Specifically, without material labels, we pool training-view foreground pixels and

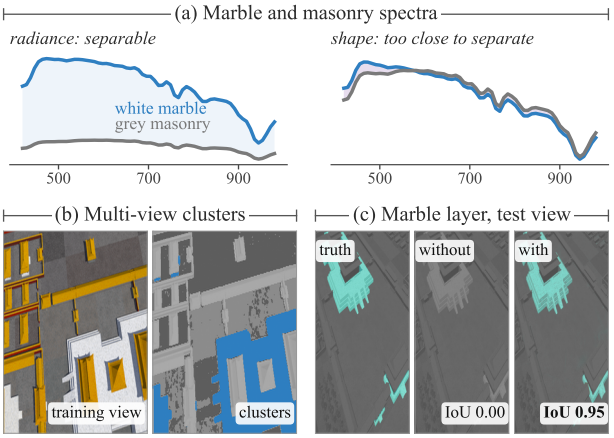


Figure 5. **Appearance regularization separates similar materials.** Forbidden City: (a) white marble is four times brighter than grey masonry but only 3.8° apart in spectral shape, allowing one scaled endmember to explain both. (b) Shape-and-brightness clustering pools pixels across training views. (c) Cluster assignments regularize abundances to keep marble and masonry distinct.

describe each 3×3-smoothed spectrum $\tilde{\mathbf{H}}(\mathbf{r})$ by its shape and brightness:

$$\mathbf{q}_\mathbf{r} = \left[\frac{\tilde{\mathbf{H}}(\mathbf{r})}{\|\tilde{\mathbf{H}}(\mathbf{r})\|_2},\ \kappa\, b(\tilde{\mathbf{H}}(\mathbf{r}))\right] \in \mathbb{R}^{L+1}, \quad (8)$$

where b is the log norm of the spectrum, standardized robustly over the training views and clipped. The bounded weight κ makes brightness secondary to shape while separating nearly parallel spectra. Farthest-point seeding and k-means group the descriptors into K clusters; as in clustering-based self-supervision [7], cluster indices provide pixel pseudo-labels $y_\mathbf{r} \in \{1, \ldots, K\}$, and cluster centres initialize the corresponding endmembers.

Persistent assignment regularization. We retain the initial assignments during optimization by penalizing low rendered abundance for each pixel's assigned endmember:

$$\mathcal{L}_p = -\frac{1}{|\mathcal{B}_p|}\sum_{\mathbf{r}\in\mathcal{B}_p} \log \hat{A}_{y_\mathbf{r}}(\mathbf{r}), \quad (9)$$

where $\hat{A}_{y_\mathbf{r}}(\mathbf{r})$ is entry $y_\mathbf{r}$ of $\hat{\mathbf{A}}(\mathbf{r})$ and $\mathcal{B}_p$ holds the pseudo-labelled pixels of the batch whose rays hit the scene. Because Gaussians are shared across views, they couple these pixel constraints, lifting 2D clusters into 3D assignments [76, 90, 95]. Thus, the loss guides assignments while spectra and intensities adapt. For the first 3k updates, endmembers remain fixed while the assignment weight decays towards the non-zero floor defined in Sec. 3.4. This floor retains guidance during joint optimization.

3.4. Optimization

Stage 1 fits the anchor and refines geometry and features on RGB with $\mathcal{L}_{\mathrm{RGB}} = 0.8\mathcal{L}_1 + 0.2\mathcal{L}_{\text{D-SSIM}}$. With this initialization, Stage 2 freezes geometry and updates features, de-

Table 2. **Spectral reconstruction and material identification on SWR-Bench.** SWR leads in mIoU and endmember accuracy across all worlds. Lower SAM need not imply better identification (3DGS on Market and Study). Rare: classes below 3 % of test pixels; Spectra: measured specimens; EM: class-matched endmember angle in degrees.

					UnMix-NeRF [64]				3DGS [38] + VCA / NNLS				Spectral World Renderer			
Scene	Setting	Classes	Rare	Spectra	PSNR↑	SAM↓	mIoU↑	EM↓	PSNR↑	SAM↓	mIoU↑	EM↓	PSNR↑	SAM↓	mIoU↑	EM↓
Forbidden City	outdoor	9	3	47	26.13	7.37	0.19	27.4	25.03	5.05	0.17	30.7	28.37	4.45	0.55	4.4
Market	outdoor	6	0	29	31.87	3.61	0.13	30.9	30.02	1.85	0.24	26.9	33.63	1.95	0.78	2.4
Botanical	outdoor	6	0	29	30.24	5.99	0.10	38.8	29.57	3.12	0.22	27.1	32.88	2.97	0.71	2.4
NVIDIA HQ	outdoor	8	2	27	18.29	14.98	0.08	43.0	22.27	12.31	0.13	24.4	25.30	8.25	0.31	3.7
Study	indoor	5	2	5	33.78	2.81	0.20	22.9	32.53	1.52	0.30	14.9	33.14	2.01	0.52	0.9
Karst	outdoor	5	2	10	26.91	3.29	0.38	25.8	27.19	2.26	0.35	28.8	28.10	2.15	0.73	2.9
City	outdoor	11	5	56	27.00	4.40	0.10	30.4	25.61	4.21	0.21	29.1	28.18	3.53	0.52	3.1
Office	indoor	8	3	35	25.33	6.65	0.11	18.8	25.01	5.03	0.17	41.3	27.54	4.31	0.53	3.3

coder and endmembers using a uniform pixel batch $\mathcal{B}$ from one camera:

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{\text{rec}} + \lambda_{\text{SAM}}\mathcal{L}_{\text{SAM}} + \lambda_p(t)\mathcal{L}_p,\\ \mathcal{L}_{\text{rec}} &= \tfrac{1}{|\mathcal{B}|L}\textstyle\sum_{\mathbf{r}\in\mathcal{B}} \|\hat{\mathbf{H}}(\mathbf{r}) - \mathbf{H}(\mathbf{r})\|_1,\\ \mathcal{L}_{\text{SAM}} &= \tfrac{1}{|\mathcal{B}_+|}\textstyle\sum_{\mathbf{r}\in\mathcal{B}_+} \text{SAM}(\hat{\mathbf{H}}(\mathbf{r}), \mathbf{H}(\mathbf{r})),\end{aligned} \tag{10}$$

The reconstruction term fits band values, SAM penalizes spectral-shape error and $\mathcal{L}_p$ guides assignments. Here $\mathcal{B}_+ \subseteq \mathcal{B}$ contains rays with non-zero target spectra, SAM is in radians, $\lambda_{\text{SAM}} = .05$ and $\lambda_p(t) = \max(.005, .05(1 - t/3000))$ at update t.

4. Experiments

We test whether spectral fidelity, dictionary size or RGB appearance alone explains material identification.

Data and baselines. SWR-Bench has eight indoor and outdoor worlds of known composition. For material identification, we compare UnMix-NeRF [64] and an L-channel 3DGS [38] unmixed afterwards with VCA [60] endmembers from training-view renders and NNLS [46]. For spectral fidelity, we additionally compare HyperGS [83] and DD-HGS [59]. Real data span 21 to 204 bands: four multispectral NeSpoF scenes [41], the 204-band HSI-SC apple [45] and five HS-NeRF scenes [9].

Evaluation protocol. Methods train on noisy 480×270 views; PSNR and the spectral angle SAM [44] (mean and 95th percentile, in degrees) are evaluated against clean 960×540 foreground ground truth. For mIoU, labelled training pixels map each endmember to the most frequent class among pixels it dominates; with this map frozen, each test pixel takes the class of its dominant endmember. Appendix A defines EM using minimum-total-angle matching between pure-pixel class means and endmembers, with equal weight for all material classes.

Implementation details. Table S4 lists the training settings of all methods. Validation mIoU selects the endmember count K per world from the class count, 16, 24 and 32: K=24 for Market, NVIDIA HQ and Study, K=32 elsewhere. Real captures use K=16; unmixing baselines on SWR-Bench use the known class count.

Table 3. **RGB geometry and feature refinement on Market.** Δ: refined minus unrefined. PSNR in dB; SAM in degrees.

RGB refinement	mIoU↑	PSNR↑	SAM↓
Without	0.66	30.73	3.20
With	**0.78**	**33.63**	**1.95**
Δ	+0.12	+2.90	−1.25

Table 4. **Training and rendering cost.** City: all four training stages. FPS includes spectra and labels at $960 \times 540 \times 64$. SWR: fused fp16, weights $\geq 10^{-4}$ (99.97 % label agreement); baseline methods use their reference implementations.

	City	FPS↑ at $960 \times 540 \times 64$				
Method	train↓	Market	Bot.	Study	Karst	Mean
UnMix-NeRF	24 min 17 s	0.18	0.21	0.21	0.12	0.18
3DGS+VCA/NNLS	85 min 2 s	2.53	2.51	4.70	4.73	3.6
SWR	**6 min 37 s**	**36.5**	**39.3**	**36.3**	**27.4**	**34.9**

4.1. Reconstruction and Material Identification

Identification and dictionary capacity. SWR reaches 0.58 mean mIoU, versus 0.16 / 0.22 for UnMix-NeRF / 3DGS+VCA/NNLS, with endmember errors of 0.9–4.4° versus at least 14.9°, as Table 2 shows. As Table S5 shows, matching K=24 nearly closes the gap on Study but leaves 3DGS+VCA/NNLS behind elsewhere.

Fidelity–identification gap. As shown in Table 2 and Figure 6, 3DGS has lower SAM yet far lower mIoU on Market and Study. For comparison, Appendix A reports 0.84–0.94 mIoU for measured-specimen NNLS on clean radiance, falling to 0.32–0.77 with noise. This reference separates ambiguity in the observations from errors introduced by learning the dictionary and assignments.

4.2. Representation and Assignment Controls

RGB feature refinement. RGB refinement improves Market by 2.9 dB and 0.12 mIoU while lowering SAM by 1.25°, as Table 3 shows. Geometry and feature layout are refined before spectral fitting; this stage takes 36.5 s on City, as Table S9 reports.

Spatial detail and shading. Without appearance regularization, one feature per Gaussian costs 1.6 dB and lowers

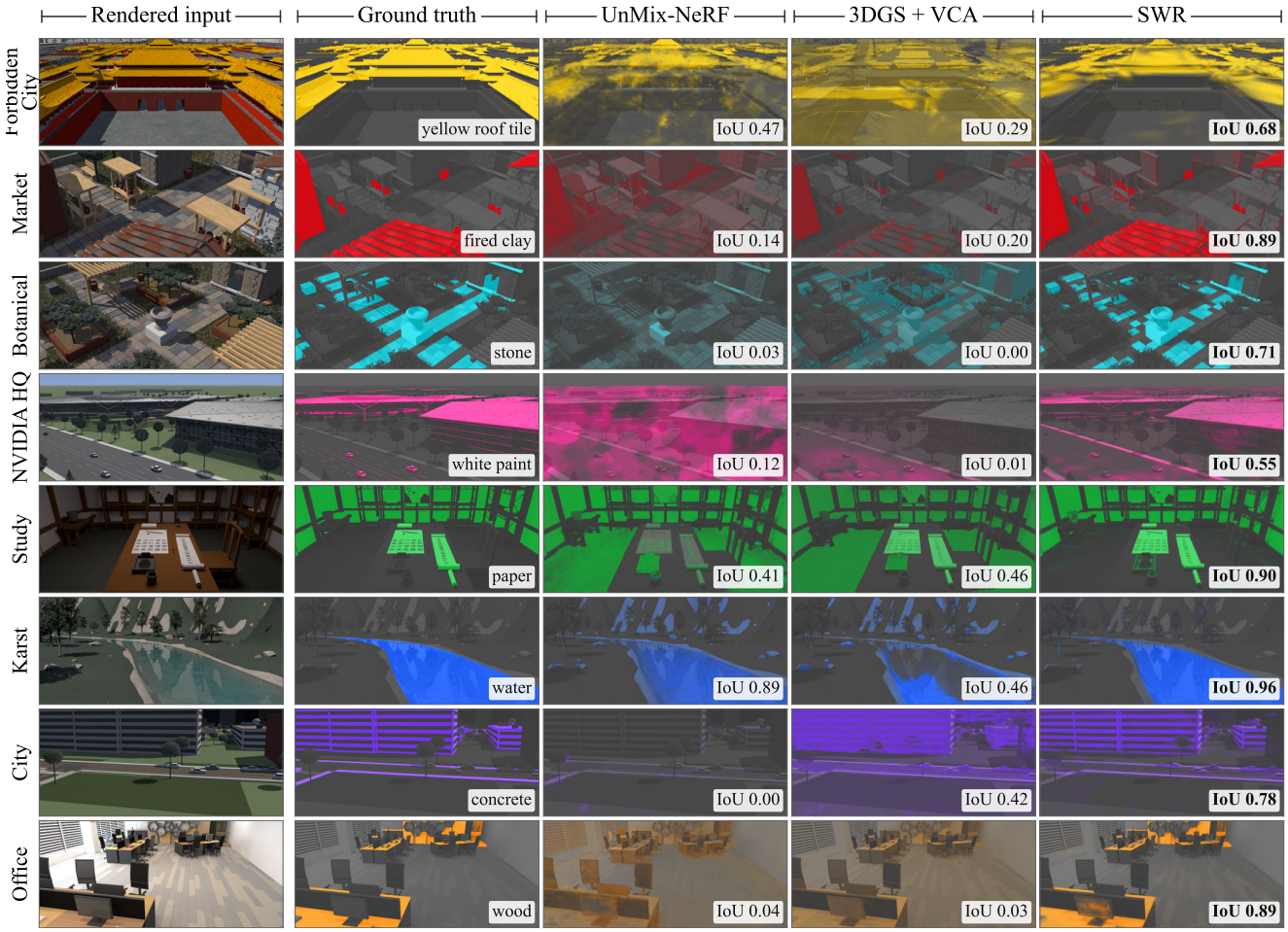


Figure 6. **Material abundance maps across eight worlds.** Each row shows one material in a test view, tinted by its abundance. All methods map at least one endmember to that material. IoU thresholds the class-summed abundance at 0.5 against the true mask. Baselines spread abundance onto unrelated surfaces. Models match Table 2.

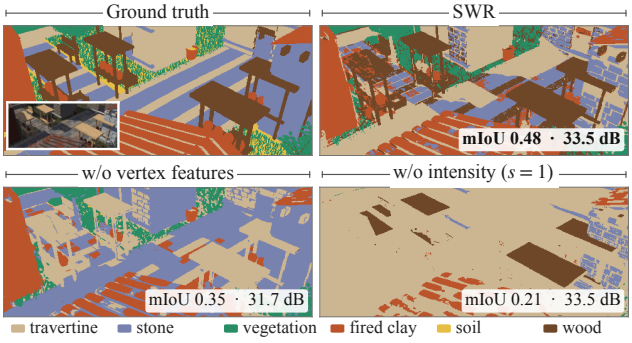


Figure 7. **Effects of decoder design.** Market test-view material maps, without appearance regularization ($\mathcal{L}_p$). Scores are view-specific mIoU and PSNR; inset: input image.

Market mIoU from 0.47 to 0.34. Likewise, fixed intensity preserves PSNR but lowers mIoU to 0.22 in Figure 7, showing how mixtures can absorb shading.

Where mixtures are decoded. Deferred decoding also raises Market PSNR while lowering mIoU from 0.47 to 0.37 in Table S6. Unlike this per-pixel prediction, intersection decoding retains primitive mixtures for cross-view queries.

Table 5. **Spectral reconstruction on real datasets.** Baseline: PSNR-best of 3DGS, HyperGS and DD-HGS. Means: 4 NeSpoF / 5 HS-NeRF scenes. L: bands; p95: 95th-percentile SAM. Angles in degrees; bold: best per dataset.

Dataset	L	Method	PSNR↑	SAM↓	p95↓
NeSpoF	21	HyperGS	28.36	4.58	9.44
		SWR	**29.39**	**3.86**	**8.04**
Apple	204	HyperGS	27.95	1.64	4.88
		SWR	**30.67**	**1.35**	**3.87**
HS-NeRF	128/141	3DGS	26.32	**5.71**	**16.75**
		SWR	**26.51**	5.97	17.31

Balancing Reconstruction and Identification

In Table 2, SWR leads the unmixing pipelines in both PSNR and mIoU on seven worlds, with highest mIoU and second-best PSNR on Study. Table S6 provides supporting decoder controls: higher PSNR can accompany lower material mIoU.

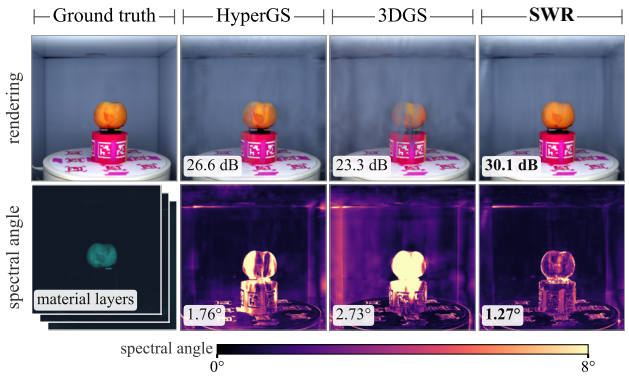


Figure 8. **Reconstruction of the 204-band HSI-SC apple.** Top: test-view renderings and PSNR. Bottom: per-pixel spectral angle and its mean. Inset: SWR material abundance maps.

Appearance and material allocation. With $K=24$ and seed 42 fixed, appearance guidance raises Market mIoU from 0.505 to 0.784 but lowers Study from 0.588 to 0.521, as Table S5 shows. However, appearance clusters can favour frequent-material illumination modes; Figure S1 shows rare classes that remain merged by SWR.

4.3. Real Captures and Efficiency

Spectral reconstruction. Table 5 shows the highest dataset-mean PSNR for SWR on each source, with its largest margin on the apple in Figure 8. On HS-NeRF, however, 3DGS has lower mean and tail angles. Thus, real captures assess spectral fidelity, while SWR-Bench provides composition truth for material identification.

Efficiency. Table 4 includes all four SWR training stages: 6 min 37 s on City, versus 24 min for UnMix-NeRF and 85 min for 3DGS+VCA/NNLS. After fitting, the optimized renderer averages 34.9 joint spectral and label views/s.

4.4. Multi-View Queries Beyond RGB

Resolving Approximate RGB Metamerism

City illustrates approximate RGB metamerism: mineral leaves differ from foliage by 2° in RGB-vector angle but 42° across HSI bands. SWR exploits this spectral distinction to separate the materials and reuse their selection across viewpoints.

From image labels to surface queries. Image-space hyperspectral classification and clustering produce pixel maps [10, 29, 55, 82, 93]. Here, multi-view HSI trains shared material features on a Gaussian scene, making spectral distinctions queryable from novel cameras. Names use training-view labels, whereas clicks and spectra need no class name. On shared candidates, RGB retrieves eleven ordinary trees and the mineral query none, as Figure 9(a) shows.

Queries under known illumination. Using known sun/sky irradiance, shadowed clicks achieve 77 % mean recall on sunlit regions across four tested worlds, as Figure 9(d) shows; Appendix H explains the model.

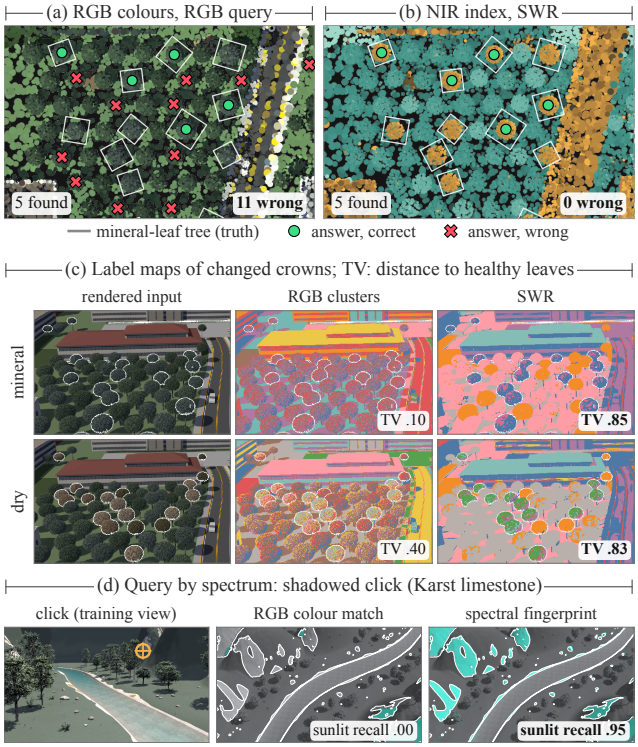


Figure 9. **Material queries beyond RGB.** City: mineral leaves replace the boxed crowns. (a,b) RGB retrieves ordinary trees as well; spectral queries isolate the mineral. Counts cover the type-query instances reported in Table S11. (c) SWR separates mineral and dry crowns from healthy foliage. (d) With known irradiance, a shadowed limestone query retrieves sunlit rock in this Karst example; recall covers all test views.

Query cost. On a fitted scene, 960×540 selection rasterization takes 3–4 ms on an RTX 3090 after Gaussian aggregation. This timing excludes scene fitting.

5. Conclusion

Spectral World Renderer makes spectral distinctions beyond RGB accessible as queryable 3D material structure. Compact Gaussian features jointly support spectral reconstruction and unmixing, while pre-compositing decoding retains material mixtures on explicit primitives. Consequently, a spectral selection can be reused across cameras instead of recomputed as independent image labels.

SWR achieves the highest mIoU across eight worlds and the highest dataset-mean PSNR on each real dataset evaluated. On City, fitting is 3.7–12.9× faster than the unmixing baselines, supporting efficient scene-level material analysis. Executable worlds with known composition, in turn, provide a reproducible basis for evaluating material identification alongside spectral reconstruction.

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Supplementary Material

Contents. Figures, tables and equations carry the prefix S; unprefixed numbers refer to the main paper.

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A. SWR-Bench Worlds and Protocol

Scene construction. Each world is defined by an executable scene program specifying geometry, material assignments, illumination and cameras. Reference photographs, maps and text guide the construction. Spectral specimens remain fixed on surfaces across viewpoints, ensuring a common material identity for all observations. Table S1 summarizes the scene configurations and camera splits.

Shared samples. Spectral images, RGB and ground truth share 16 sub-pixel samples per pixel, so labels and area proportions describe the first-hit surfaces. Radiance reflected or transmitted by water and glazing is therefore attributed to the water or glazing class, and interreflected light to the surface on which it is observed; transmitted and interreflected light is stored as separate contributions.

Rendering. All worlds share 64 Gaussian bands centred from 420 to 980 nm (20 nm full width at half maximum) at 960×540. Outdoor worlds use Lambertian surfaces under ASTM G173 sunlight and skylight [1]; NVIDIA HQ adds glossy highlights, glass reflection and an analytic sky, and Karst adds refracting water with pure-water absorption. The Study and Office are path traced in Mitsuba [37] with interreflection, sampling reflectance at the band centres. RGB is noiseless: CIE 1931 colour matching gives XYZ, which is adapted to D65 by the Bradford transform (70 % partial adaptation indoors) and converted to sRGB with one fixed exposure per world.

Noise. Every world is released clean and noisy, with Poisson and read noise added per band to the 960×540 radiance h. With N_e electrons at the reference radiance L_{ref} and read noise σ_r as listed in Table S1, the observation is $\tilde{h} = (L_{\mathrm{ref}}/N_e)\,[\mathrm{Poisson}(N_e h/L_{\mathrm{ref}}) + \mathcal{N}(0, \sigma_r^2)]$. Outdoors, L_{ref} is the maximum clean radiance over all training pixels and bands, and negative values are clipped to zero; in the Study and Office, it is the 99.5th percentile of the first training view, excluding emitters, without clipping.

Evaluation protocol. Training views are 2×2 box averages of the noisy cubes, normalized by L_{ref}, at 480×270; evaluation uses clean 960×540 foreground radiance. All methods share camera splits, classes and the stereo point cloud. Ground-truth consistency is verified by full primary-ray coverage, at least 99.9 % agreement between instance and material maps, and exact recomposition of outdoor radiance from the stored coefficients.

Classes and City variants. Finer labels are merged into one class when they play the same material role with similar spectra, for example the marbles and granite of Market and Botanical, the near-neutral grey paving and brick of Forbidden City, and bark and soil; spectrally distinct labels remain separate even when rare, such as painted eaves and yellow road paint, whereas white road paint (0.3 % of City pixels) is merged into asphalt. Four City variants change one factor each: foliage-coloured mineral leaves (*metamer*), dry leaves on 87 trees (*stress*), and a morning or afternoon sun. Every object keeps its instance and spectral sample.

Metrics. PSNR and spectral angle are computed against clean radiance on the ground-truth support. For non-zero spectra $\mathbf{x}, \mathbf{y}$, $\mathrm{SAM}(\mathbf{x}, \mathbf{y}) = \arccos(\mathbf{x}^\top \mathbf{y}/(\|\mathbf{x}\|_2 \|\mathbf{y}\|_2))$; reported angles use degrees.

- *Labels.* Endmembers are mapped many-to-one to classes by majority vote on training views. With the endmember-to-class map π frozen, each test pixel takes the class of its dominant endmember, $c = \pi(\arg\max_k \hat{A}_k)$, not the class of largest summed abundance; endmembers that dominate no training pixel stay unassigned.
- *Endmember angle (EM).* Each class is paired with a distinct endmember by the linear-sum (Hungarian) assignment of minimum total angle to the mean clean spectrum of its pure training pixels (area proportion at least 0.9). With $K \geq C$, the remaining $K - C$ endmembers do not enter EM, and the map used for mIoU plays no role.
- *True-spectrum reference (GT dict.).* Every second test pixel along each axis is unmixed by non-negative least squares against all measured specimens of the world, each under sunlight and skylight, and labelled by the largest class-summed contribution; Karst is scored on its opaque classes. The reference reaches 0.84–0.94 mIoU on clean radiance but 0.32–0.77 under noise, and gives the true-spectrum IoU of Figure S1. It is undefined for the path-traced Study and Office, which include interreflection, and for NVIDIA HQ, whose glass reflections and haze fall outside the first-surface dictionary.

Rare-material identification. Figure S1 relates per-class IoU to pixel frequency. Known-specimen NNLS separates many rare classes merged by SWR, indicating that dictionary learning and material allocation contribute to the errors beyond ambiguity in the observations.

Cross-view evaluation. Figures S8 and S9 illustrate the hyperspectral observations. Figures S10–S13 compare ground truth and SWR predictions at three test cameras per world,

Table S1. **The eight worlds of SWR-Bench.** Classes and spectra: material classes and distinct measurements (the total deduplicates measurements across worlds) listed in Table S2. Views are split by camera into training, validation and test. Noise: electrons N_e at the reference radiance and read noise σ_r. Obs.: PSNR of noisy against clean test views. City variants share these cameras and splits.

						Views				
World	Content	Cameras	Transport	Classes	Spectra	Train	Val	Test	N_e / σ_r	Obs. (dB)
Forbidden City	palace: golden roofs, courtyards, water	3 orbits, 45–430 m	two-term	9	47	135	21	24	5,000 / 2	45.4
Market	market square: stalls, pottery, walls	3 rings, inward and downward	two-term	6	29	135	21	24	5,000 / 2	45.6
Botanical	botanical garden: planters, trees, pergolas	3 rings, inward and downward	two-term	6	29	135	21	24	5,000 / 2	46.4
NVIDIA HQ	offices: glazed facades, 300 trees	spiral, facade, campus routes	extended	8	27	180	28	32	5,000 / 2	46.7
Study	scholar's study: shelves, paper, ink, ceramics	eye-level and elevated orbits	path traced	5	5	64	16	16	3,000 / 3	44.8
Karst	river valley: limestone cliffs, clear water	river banks, cliff, high loop	two-term + water	5	10	128	32	32	5,000 / 2	44.3
City	campus block: roads, 297 trees, vehicles	4 orbits, 16–95 m	two-term	11	56	180	28	32	5,000 / 2	45.2
Office	office: 35 LED downlights, daylight	eye-level and elevated loops	path traced	8	35	64	16	16	3,000 / 3	46.0
Total					**125**	**1,021**	**183**	**200**		
City variants	metamer, stress, morning, afternoon	as City	two-term	11–12	56–60	as City			as City	–

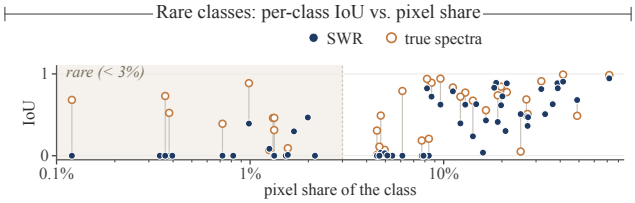


Figure S1. **Per-class IoU and class frequency.** Per-class IoU against pixel share over the eight worlds, with K equal to the class count. NNLS with all measured specimens (GT dict.) separates most rare classes that Spectral World Renderer merges.

using the main-paper models and fixed training-view class maps. Camera separation from a reference view determines the examples independently of prediction quality.

B. Spectral Sources and Validation

Sources and selection. Reflectance comes from ECOSTRESS [54] and the USGS Spectral Library 7 [43], as shown in Table S2. A record is accepted only if it is a reflectance measurement that covers 400–1000 nm with gaps of at most 20 nm and values in $[0, 1]$, and if its metadata match the assigned material class, such as photosynthetic grass for foliage or bark for trunks. Accepted records are interpolated on a 1 nm grid without smoothing or extrapolation and integrated against 64 Gaussian bands of 20 nm full width at half maximum, truncated at ± 20 nm. Several specimens per class and separate spectra under sunlight and skylight give each material a range of appearances, as shown in Figure 3 of the main paper.

Realism. Rendered natural materials agree closely with independent measurements of the same type, excluding rendered records, as reported in Table S3: 37 of 40 eligible natural-material assignments lie within 3° of a reference, and all 21 vegetation spectra pass red-edge and NDVI checks [30, 84]. The dry and mineral substitutes in the City variants lack these vegetation signatures. Man-made materials are more varied, and pigment proxies for paints deviate most.

Proxies. Some objects use a measured specimen of a related material, such as pigment powders for paints and road markings, cotton for paper, or granite for an inkstone.

C. Implementation Details

RGB anchor. Table S4 lists the training settings of all methods. The anchor uses multi-view density control [72], which clones and splits Gaussians that cover many high-error pixels and prunes by opacity and multi-view importance. After fitting, colours are discarded and the rasterizer's screen-space variance is folded into the scales,

$$\boldsymbol{\sigma}_i^2 \leftarrow \boldsymbol{\sigma}_i^2 + 0.3\,\mathrm{med}_v(z_{iv}/f_v)^2, \tag{S1}$$

The square and update are componentwise. Here z_{iv} is the depth of Gaussian i in training camera v, f_v the focal length in pixels of the 480×270 training views, and the median runs over the training cameras that see the Gaussian; the term transfers the $0.3\,\mathrm{px}^2$ low-pass variance of the anchor rasterizer to the 3D kernel. The harmonic features are then refined for 5k steps without densification. The RGB objective combines mean absolute error $\mathcal{L}_1$ and structural dissimilarity $\mathcal{L}_{\text{D-SSIM}}$. Each Gaussian stores three position coordinates, a four-value rotation quaternion, three scales and one opacity, plus four 12-dimensional feature vectors. The encoding $[\sin \mathbf{f}; \cos \mathbf{f}]$ concatenates elementwise sine and cosine.

Vertex interpolation. The canonical tetrahedron is regular with inradius 1 in the frame of the Gaussian scaled by $\boldsymbol{\sigma}_i$, with vertices $(\pm\sqrt{6}, -\sqrt{2}, -1)$, $(0, 2\sqrt{2}, -1)$ and $(0, 0, 3)$. The barycentric coordinates of the point of maximal response are affine in its position and are not clamped, so they extrapolate linearly outside the tetrahedron.

Appearance clusters. With the sampling settings in Table S4, smoothed pixel spectra $\mathbf{x} = \tilde{\mathbf{H}}(\mathbf{r})$ are described by $[\mathbf{u}, \kappa b]$, with the unit spectrum $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|$ and the standardized log norm $b = \mathrm{clip}\big((\log \|\mathbf{x}\| -$

Table S2. **Materials and measured spectral sources in SWR-Bench.** Each row identifies the material classes in a world and the number of measurements drawn from each spectral library. Counts are per world; numbers after material names count their measurements.

		Measured spectra			
World	Classes	ECOSTRESS	USGS	Total	Material classes (measurements)
Forbidden City	9	27	20	47	yellow glazed tile (5); vermilion paint (3); vegetation (8); white marble (4); grey masonry (9); painted eaves (4); soil/bark (7); water (3); distant haze (4)
Market	6	29	0	29	travertine (1); stone (white marble, pink marble, granite) (7); vegetation (8); fired clay (4); sandy soil (4); wood/bark (5)
Botanical	6	29	0	29	travertine (1); stone (white marble, pink marble, granite) (7); vegetation (8); fired clay (4); sandy soil (4); wood/bark (5)
NVIDIA HQ	8	21	6	27	white coating/marking (3); glazing (1); vegetation (8); dark panels/frames (2); concrete (3); asphalt (2); soil/bark (7); wood (1)
Study	5	3	2	5	aged wood (1); paper (1); ink (1); inkstone (1); ceramic (1)
Karst	5	10	0	10	limestone (2); vegetation (5); soil (1); wood/bark (2); water (absorption model)
City	11	33	23	56	concrete/masonry (7); soil/bark (6); vegetation (7); fired clay (8); asphalt/markings (4); yellow road paint (2); dark glazing (2); painted facade (6); metal roofing (4); vehicle (8); dark backdrop (2)
Office	8	11	24	35	light oak (6); dark laminate (5); textile (5); grey vinyl (3); white paint/plastic (6); vegetation (4); metal (4); display (2)
Controlled City variants; all other material assignments are unchanged.					
City-metamer	12	35	24	59	City materials, plus a mineral substitute (3)
City-stress	11	36	24	60	City materials, with vegetation expanded from 7 to 11 measurements

Class grouping. Evaluation merges 11 labels into 9 in Forbidden City, 8 into 6 in Market and Botanical, 14 into 11 in City, and 15 into 12 in City-metamer. *Illumination.* Outdoor worlds use G173 sunlight and skylight; Study uses a 5000 K panel, and Office uses LED-B3 lighting with G173 daylight. City variants retain the City geometry, cameras and sensor. *Material proxies.* Forbidden City uses measured pigments for tiles and paints, reflectance proxies for water and haze; Karst water is defined by absorption and a Fresnel interface, without a reflectance record.

$m)/\max(0.1,\, 1.4826\,\mathrm{MAD}), -4, 4\big)$, where m and MAD are the median and the median absolute deviation of $\log\|\mathbf{x}\|$. Here $\|\cdot\|$ denotes the Euclidean norm. The weight $\kappa = \mathrm{clip}(0.5\,\mathrm{med}\|\mathbf{u} - \bar{\mathbf{u}}\|, 0.05, 0.2)$, with $\bar{\mathbf{u}}$ the normalized mean of $\mathbf{u}$, adapts to each scene. Farthest-point seeding and 20 Lloyd steps on the unit sphere give K clusters. Endmember k is parametrized as $\bar{\mathbf{e}}_k = \mathrm{sigmoid}(\boldsymbol{\eta}_k)/\|\,\mathrm{sigmoid}(\boldsymbol{\eta}_k)\|_2$, and $\boldsymbol{\eta}_k = \mathrm{logit}(\mathbf{c}_k)$ is initialized from the unit spectral part $\mathbf{c}_k$ of cluster centre k, clipped to $[10^{-5}, 1 - 10^{-5}]$.

Endmember capacity and material separation. A material can span several spectral appearances, so the endmember count need not equal the number of material classes. Table S5(a) shows that a larger dictionary improves identification over the class-count setting in every world. The validation-selected size is also the best test candidate in seven of eight worlds.

Panel (b) compares methods at the same $K = 24$. SWR remains ahead of 3DGS+VCA/NNLS, with a small gap on Study, so capacity alone does not explain the identification advantage. Appearance guidance improves six worlds but reduces mIoU on Study and Karst, so allocation guidance and dictionary size play distinct roles.

D. Decoder Analysis

Reconstruction and material assignments. Table S6 compares decoder variants without appearance regularization, averaged over seeds 42 and 43. Replacing the mixture decoder with a free spectral decoder improves PSNR by at most 0.07 dB, whereas replacing the spatially varying features with one feature per Gaussian costs 0.4–1.6 dB. Reconstruction quality therefore depends on the feature field.

Material assignments respond differently. Removing harmonic encoding or independent intensity preserves much of Market's PSNR but reduces mIoU, as mixtures absorb shading. Deferred decoding raises Market PSNR from 33.72 to 34.37 dB while lowering mIoU from 0.47 to 0.37. This comparison illustrates why spectral fidelity alone cannot establish correct material decomposition. Pre-compositing decoding also retains a mixture at each Gaussian intersection for direct primitive queries.

Initialization and assignment guidance. On Market at K=24, appearance regularization alone reaches 0.771 mIoU and cluster initialization alone reaches 0.732. Their combination reaches 0.784, compared with 0.505 without either. The two components therefore provide complemen-

Table S3. **Validation of the assigned reflectance spectra.** Physical-range checks test whether reflectance stays in $[0, 1]$. Independent-reference checks compare each eligible spectrum with other measurements of its material type. Leaf checks: red edge and NDVI.

Scope	Spectra	Physical range	Independent reference agreement		Vegetation
		Valid / tested	Within 3° / eligible	Median angle (degrees)	Leaf checks passed
Forbidden City	47	47/47	22/24	1.28	8/8
Market	29	29/29	17/21	2.01	8/8
Botanical	29	29/29	17/21	2.01	8/8
NVIDIA HQ	27	27/27	17/19	1.35	8/8
Study	5	5/5	1/2	6.09	–
Karst	10	10/10	9/9	0.60	5/5
City	56	56/56	20/30	2.07	7/7
Office	35	35/35	8/18	3.46	4/4
Summary across distinct measurement–material pairs					
Six outdoor worlds	119	119/119	47/61	1.47	17/17
All eight worlds	154	154/154	55/78	2.00	21/21
natural materials	45	45/45	37/40	1.25	21/21
man-made materials	109	109/109	18/38	3.46	–

An eligible spectrum has at least three independent library references of the recorded material type; the median is the nearest-reference spectral angle over eligible spectra. Proxy assignments without such references are excluded from this test. World rows count a shared measurement in every world where it appears. Summary rows count each measurement–material pair once: 154 pairs from 125 distinct spectra. Natural materials comprise vegetation, bark, soil and rock. Leaf checks are not applicable to Study or man-made materials.

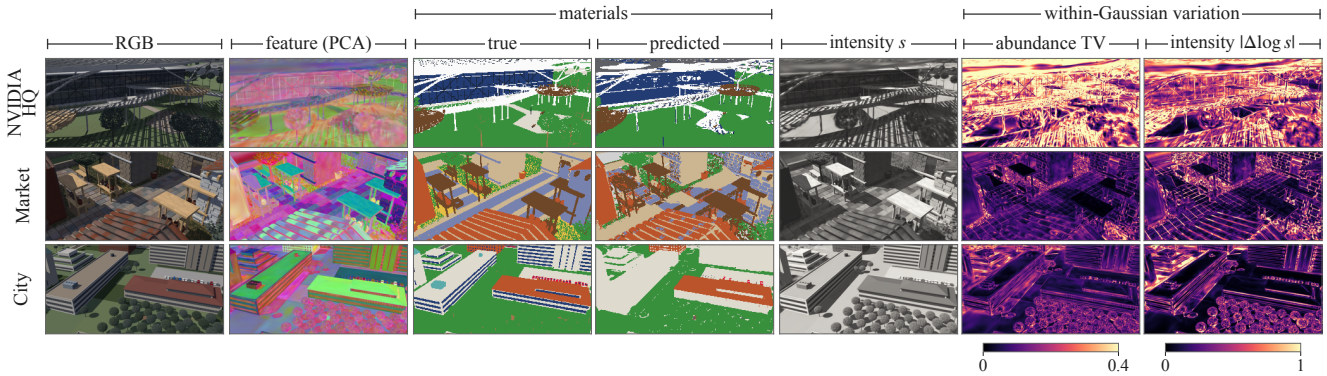


Figure S2. **Harmonic features and their decoding.** Columns show RGB, the per-hit feature (PCA), true and predicted materials, intensity, and the variation of mixture and intensity within each Gaussian against its mean feature in the same viewing direction (fixed scales).

tary guidance for material allocation. The adaptive brightness weight κ outperforms fixed weights of 0.05 and 0.2.

Feature structure. Projected on their principal components, the features follow both material and shading, as shown in Figure S2. Within one Gaussian, intensity varies with shading, while mixture coefficients change mainly at material boundaries, so a single primitive can span an edge.

Material structure on explicit primitives. The primitive visualizations in Figure S3 associate surface regions with their decoded material classes. Class boundaries in the rendered maps align with changes between primitives, illustrating how a material selection can be carried by the 3D representation and rendered across viewpoints.

E. Real Captures

Alignment and scene selection. HS-NeRF evaluation requires a unique correspondence among hyperspectral cubes, RGB images and COLMAP cameras. Fakeplant2 and Pinecone are excluded because their processed RGB backgrounds differ from the raw hyperspectral captures. Figures S4–S6 compare the first test view of each evaluated scene; errors concentrate at boundaries and thin structures.

Table S4. **Training configuration by stage.** All methods use the same noisy 480×270 training views, normalized by L_{ref} and clamped to $[0, 1]$, with a shared stereo point cloud. Evaluation renders at 960×540.

Setting	Configuration
RGB geometry and feature refinement	
RGB anchor	30,000 iterations
Harmonic features	5,000 refinement steps; no further densification
Appearance clustering	
Spectrum sampling	At most 2,304 spectra per view, after a 3×3 box filter
Foreground selection	Spectrum norm greater than 0.03
Spectral fitting	
Optimizer and duration	Adam, 30,000 updates
Training batch	2,048 pixels sampled from one camera per update
Learning-rate schedule	10^{-3}, with cosine decay to 10^{-5}
Endmember optimization	Fixed for the first 3,000 updates, then optimized at 10^{-3}
Assignment-loss weight	$\lambda_p(t) = \max(0.005,\ 0.05(1 - t/3000))$
Evaluation weights	Exponential moving average, decay 0.99
Decoder controls	No appearance regularization; K equals the class count on SWR-Bench, $K = 16$ on real captures
Numerical safeguards	
Abundance normalization	Denominator bounded below by 10^{-8}
Assignment-loss support	Apply $\mathcal{L}_p$ only to rays with $\sum_i w_i > 10^{-6}$
Baseline configurations	
3DGS + VCA/NNLS	64-channel 3DGS, 30,000 steps; VCA endmembers and NNLS abundances [46, 60]
UnMix-NeRF	Official implementation and training schedule, 30,000 iterations

F. Efficiency

Representation size. Per-Gaussian storage is independent of band count: 59 values for SWR, compared with 78 for HyperGS and $62+L$ for per-band fields. The shared endmember dictionary additionally requires LK values. These counts describe primitive parameters; complete model storage also includes the shared decoder and dictionary.

Computational cost. Table S9 reports 397 s for the complete City pipeline. Only appearance clustering and spectral decoding depend on K; anchor construction and RGB feature refinement are shared across dictionary sizes. Per-intersection decoding adds computation relative to a stored spectrum. The fused half-precision kernel reduces this cost through joint evaluation and negligible-weight pruning, reaching 34.9 labelled views/s, approximately ten times the 3DGS+VCA/NNLS rate in Table S10.

G. Leaf Variants and Object Queries

Altered leaves. In the City variants, Spectral World Renderer separates mineral and stressed leaves from healthy foliage better than RGB clustering: the total variation between their label distributions is 0.85 and 0.83, against at most 0.40 for RGB k-means, (main paper, Figure 9(c)).

Object-level spectral selection. Each Gaussian is assigned its mean abundance and radiance over the training views. The type "tree" denotes crown materials and a size: Gaussians of crown material above the ground are clustered with DBSCAN [15], and large clusters form instances. A measured leaf spectrum then selects instances by comparing each crown with the query and with ordinary leaves under sunlight and skylight. All query parameters are fixed from the base City configuration.

Spectral discrimination and instance coverage. The substituted mineral differs from foliage by 42° in spectral angle but only 2° in RGB angle. Table S11 separates spectral false positives from targets lost during material assignment or instance formation. Spectral queries eliminate ordinary-tree false positives in the metamer variant, while retrieval coverage remains limited by sparse crowns. Known-specimen spectra produce the same selections, indicating that the missed instances arise before spectral matching.

H. Spectral Material Queries

Illumination-aware spectral matching. The query is a spectrum: the rendered spectrum at a clicked pixel or a library reflectance $\boldsymbol{\rho}$. Each Gaussian carries its visibility-weighted mean spectrum over training rays with non-zero

Table S5. **Endmember capacity and material identification.** Panel (a) examines how dictionary size affects the regularized SWR model. Panel (b) fixes the dictionary size to separate capacity from differences in representation and appearance guidance. Scores are test mIoU.

(a) Increasing dictionary capacity

		Test mIoU by endmember count				Validation selection	
World	Classes C	$K = C$	$K = 16$	$K = 24$	$K = 32$	Selected K	Validation mIoU
Forbidden City	9	0.410	0.500	0.538	0.552	32	0.563
Market	6	0.340	0.611	0.784	0.754	24	0.783
Botanical	6	0.315	0.628	0.684	0.712	32	0.713
NVIDIA HQ	8	0.202	0.307	0.311	0.315	24	0.320
Study	5	0.402	0.516	0.521	0.517	24	0.517
Karst	5	0.504	0.562	0.561	0.725	32	0.727
City	11	0.263	0.370	0.477	0.519	32	0.507
Office	8	0.437	0.500	0.524	0.527	32	0.533

(b) Comparing material assignments at matched capacity ($K = 24$)

World	SWR with guidance	SWR without guidance	3DGS + VCA/NNLS
Forbidden City	**0.538**	0.396	0.277
Market	**0.784**	0.505	0.323
Botanical	**0.684**	0.629	0.356
NVIDIA HQ	**0.311**	0.247	0.199
Study	0.521	**0.588**	0.514
Karst	0.561	**0.694**	0.395
City	**0.477**	0.348	0.179
Office	**0.524**	0.307	0.163

C denotes material classes and K learned endmembers. Shading in (a) marks the size selected on validation views; bold in (b) marks the best test result within each world. Class maps are fitted on training views. Guidance comprises cluster initialization and assignment regularization; the control removes both. Comparisons use seed 42; Karst includes water.

Table S6. **Decoder controls** (no appearance regularization; means of seeds 42 and 43). Mix: unmixing decoder (✗: free decoder outputting spectra). Hit: per-intersection decoding (✗: deferred, per pixel). Vtx: four vertex features (✗: one per Gaussian). Harm.: harmonic encoding. Int.: separate intensity (✗: s=1). (h) is NHT.

						NeSpoF scene3		Market		
Exp.	Mix	Hit	Vtx	Harm.	Int.	PSNR↑	SAM↓	PSNR↑	SAM↓	mIoU↑
(a)	✓	✓	✓	✓	✓	29.62	4.22	33.72	1.87	**0.47**
(b)	✓	✓	✗	✓	✓	29.24	4.30	32.09	2.08	0.34
(c)	✓	✓	✓	✗	✓	29.61	4.21	33.65	1.89	0.30
(d)	✓	✗	✓	✓	✓	29.16	**4.17**	34.37	1.57	0.37
(e)	✓	✓	✓	✓	✗	29.54	4.37	33.70	2.14	0.22
(f)	✗	✓	✓	✓	–	**29.69**	4.28	33.74	1.72	–
(g)	✗	✓	✗	✓	–	29.26	4.34	32.11	1.90	–
(h)	✗	✗	✓	✓	–	29.21	**4.17**	**34.38**	**1.41**	–

Table S7. **Real captures and splits.** Test views are spread along each trajectory, about one view in seven; NeSpoF also uses every fifth training view for validation. HS-NeRF sensors: Surface Optics (SO) and BaySpec (BS). Bold: totals.

Dataset / scene	Bands	Range (nm)	Resolution	Train	Test
NeSpoF	21	450–650	512 × 428		
scene1 / 2 / 3 / 4				26 / 21 / 15 / 21	4 / 4 / 3 / 4
total				**83**	**15**
HSI-SC apple	204	397–1004	512 × 512	**56**	**8**
HS-NeRF (SO)	128	371–1044	348 × 260		
Basil / Rosemary				41 / 37	6 / 6
Harsh1 / Harsh2				42 / 20	7 / 3
HS-NeRF (BS)	141	401–1087	320 × 256		
Fakeplant1				50	8
total				**190**	**30**
All real captures				**329**	**53**

visibility, $\bar{\mathbf{h}}_i = \sum_{\mathbf{r}} w_i(\mathbf{r})\mathbf{h}_i(\mathbf{r}) / \sum_{\mathbf{r}} w_i(\mathbf{r})$. Let $\boldsymbol{\phi}_{\mathrm{d}}, \boldsymbol{\phi}_{\mathrm{s}} \in \mathbb{R}^L$ denote the direct and sky irradiance of the world integrated over the band responses; both are known for SWR-Bench, and the cone and the whitening below use them. Under a two-component sun/sky approximation, surface radiance is modelled as proportional to $\boldsymbol{\rho} \odot (\xi\boldsymbol{\phi}_{\mathrm{d}} + \boldsymbol{\phi}_{\mathrm{s}})$, where $\odot$ denotes bandwise multiplication and $\xi \in [0, 1]$ the direct-shading coefficient. We reduce this modelled variation using two comparisons:

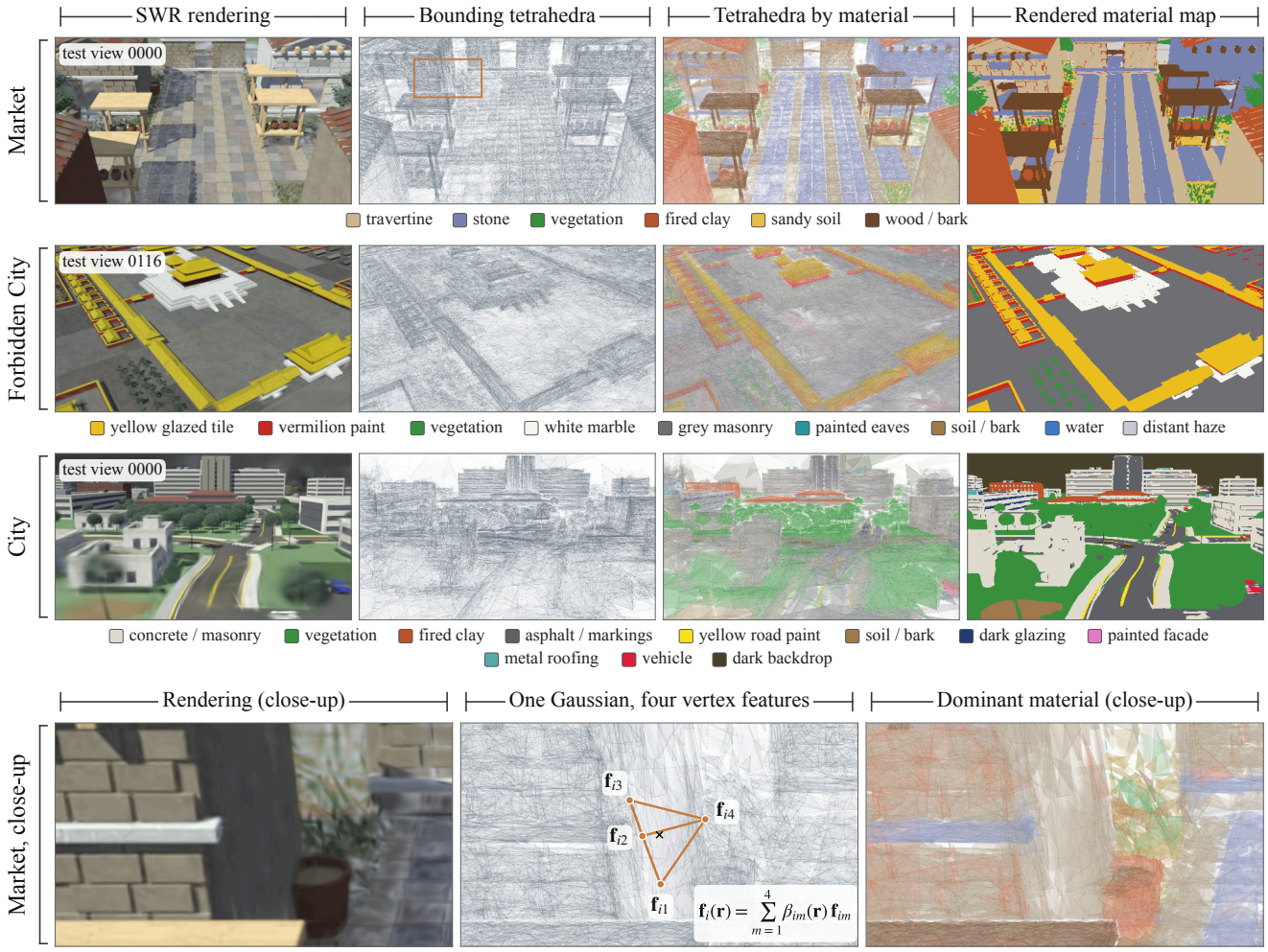


Figure S3. **Gaussian tetrahedra and decoded materials.** Tetrahedra follow each Gaussian's scale, rotation and centre; their barycentric coordinates interpolate four vertex features according to Eq. 3 of the main paper. Columns show the rendering, opacity- and visibility-weighted tetrahedra, centre-decoded classes, and material maps. The median visible Gaussian assigns more than 98 % of its centre abundance to one class. Bottom: one Market Gaussian, marked by the amber window.

- a library spectrum is compared by its angle to the non-negative cone spanned by $\boldsymbol{\rho}\odot\boldsymbol{\phi}_{\mathrm{d}}$ and $\boldsymbol{\rho}\odot\boldsymbol{\phi}_{\mathrm{s}}$, which spans the illumination mixtures in this model;
- a clicked spectrum is compared after illumination whitening: the root-mean-square (RMS) log-ratio of the two spectra once the constant vector and the two principal directions of $\log(\xi\boldsymbol{\phi}_{\mathrm{d}} + \boldsymbol{\phi}_{\mathrm{s}})$, $\xi \in [0, 1]$, are projected out.

The constant direction removes overall intensity; the two principal directions approximate sun/sky colour variation. Both use the known irradiance and assume this transport model; unknown lighting is not estimated. Rendering weights rasterize the per-Gaussian angles, and Otsu's method [62] estimates the selection threshold from the histogram of log angles.

Evaluation. Market, City, Botanical and Karst use the models of Table 2. Clicks are three sunlit and three shadowed pixels inside a single class in training views. Library queries use the assigned specimen of each class, another specimen of the class and ECOSTRESS spectra [54] that do not occur in the world. Each query is scored by AP against the class mask on all test views, and the four worlds are weighted equally. Click queries operate on rendered pixels; per-Gaussian queries differ from them by at most 0.02 AP.

Approximate RGB metamerism. The City mineral substitute differs from foliage by 2° in RGB angle but 42° in spectrum. This is an approximate camera-colour match rather than an exact CIE metameric match. A living-leaf spectrum separates the two leaf types with AUC 0.99, compared with 0.26 for RGB, and largely excludes mineral crowns, as Figure S7(c) shows. Both targets and distractors are tree crowns, reducing object-category cues in this test.



Figure S4. **Qualitative comparison on HS-NeRF test views**, the first test view of each of the five scenes. Rows show false colour (630, 550 and 460 nm) and per-pixel spectral angle (left scale). Bold marks the best scores.

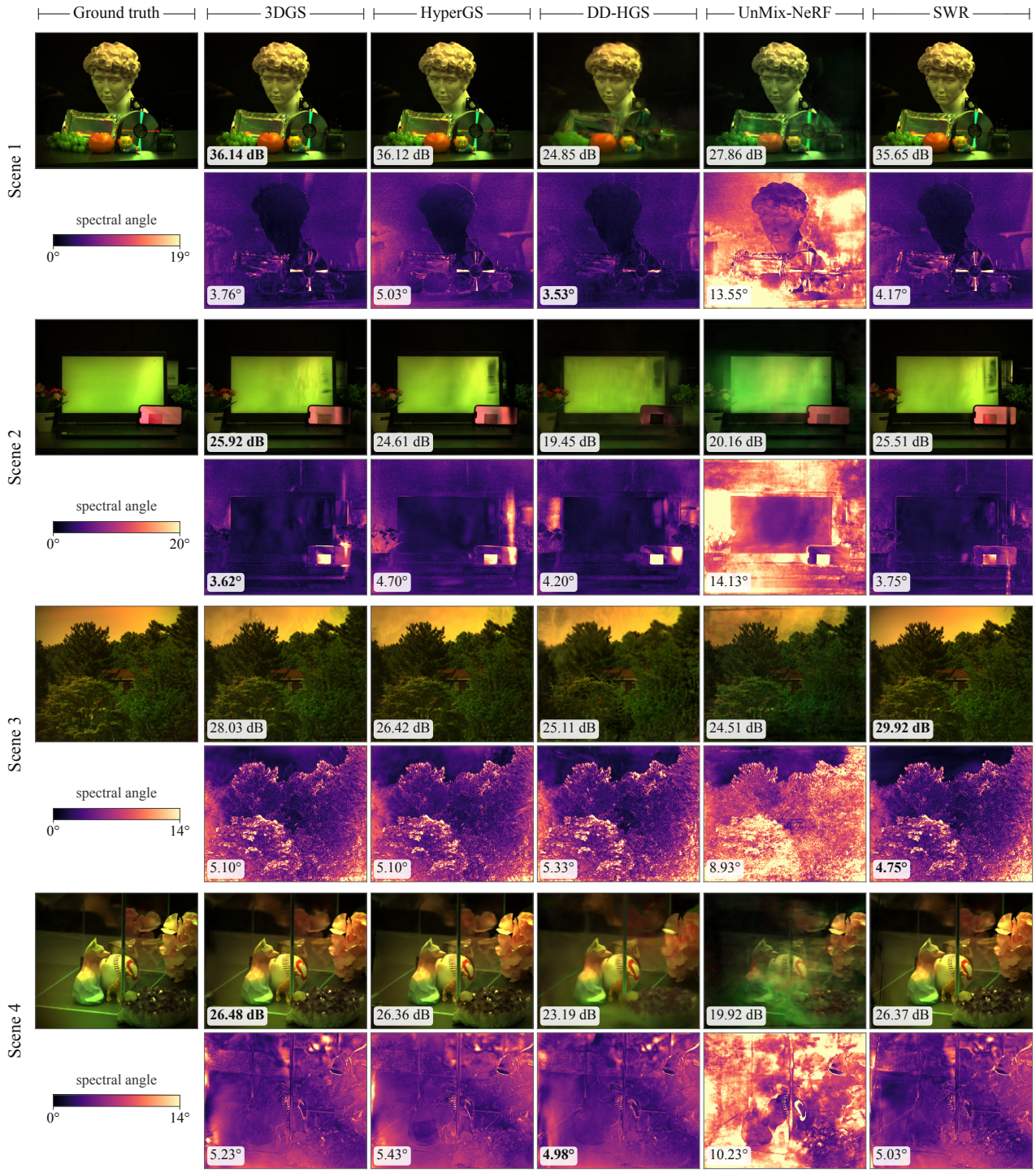


Figure S5. **Qualitative comparison on NeSpoF test views**, the first test view of each of the four scenes, as in Figure S4.

Table S8. **Spectral reconstruction on real captures, per scene** (test views; the main paper reports the means). MS/HS: measured multispectral/hyperspectral; 3DGS stores one channel per band. PSNR is higher-is-better; SAM and p95 angle are lower-is-better. Darker shading indicates better ranks; OOM denotes out-of-memory cases.

Source	Scene	Spectral World Renderer			HyperGS			DD-HGS			3DGS			UnMix-NeRF		
		PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95	PSNR	SAM	p95
MS	NeSpoF scene1	30.74	3.82	7.12	31.38	4.66	9.46	21.77	3.61	6.65	30.68	3.63	6.82	21.77	13.70	22.79
	NeSpoF scene2	28.38	3.57	8.00	26.56	4.30	9.15	21.40	3.47	7.64	27.57	3.32	7.44	21.69	14.33	22.41
	NeSpoF scene3	29.80	4.19	9.36	26.05	5.11	10.91	25.42	4.91	10.64	25.13	5.40	11.33	20.54	8.90	15.41
	NeSpoF scene4	28.64	3.85	7.66	29.44	4.27	8.24	23.09	3.44	6.82	29.07	3.68	7.41	19.36	11.56	20.40
	Mean	29.39	3.86	8.04	28.36	4.58	9.44	22.92	3.86	7.94	28.11	4.01	8.25	20.84	12.12	20.25
HS	HSI-SC apple	30.67	1.35	3.87	27.95	1.64	4.88	26.69	1.67	5.07	26.29	1.65	5.36	OOM		
HS	Basil	30.30	6.19	17.19	29.58	7.01	18.51	28.32	6.67	17.66	29.99	6.21	17.13	19.42	17.10	34.39
	Rosemary	29.21	4.30	21.48	29.05	4.60	22.35	27.70	6.96	27.18	28.57	4.43	23.31	18.67	15.82	37.50
	Harsh1	21.31	6.22	17.69	20.92	6.81	19.14	20.95	5.99	16.55	21.02	5.99	16.88	16.26	13.28	41.92
	Harsh2	24.78	6.71	17.60	23.07	6.54	16.08	22.34	5.84	14.73	23.09	6.02	15.20	15.34	21.28	51.16
	Fakeplant1	26.93	6.44	12.60	27.86	7.49	15.55	30.12	5.86	11.62	28.93	5.92	11.24	22.33	9.63	21.15
	Mean	26.51	5.97	17.31	26.10	6.49	18.33	25.89	6.26	17.55	26.32	5.71	16.75	18.40	15.42	37.22

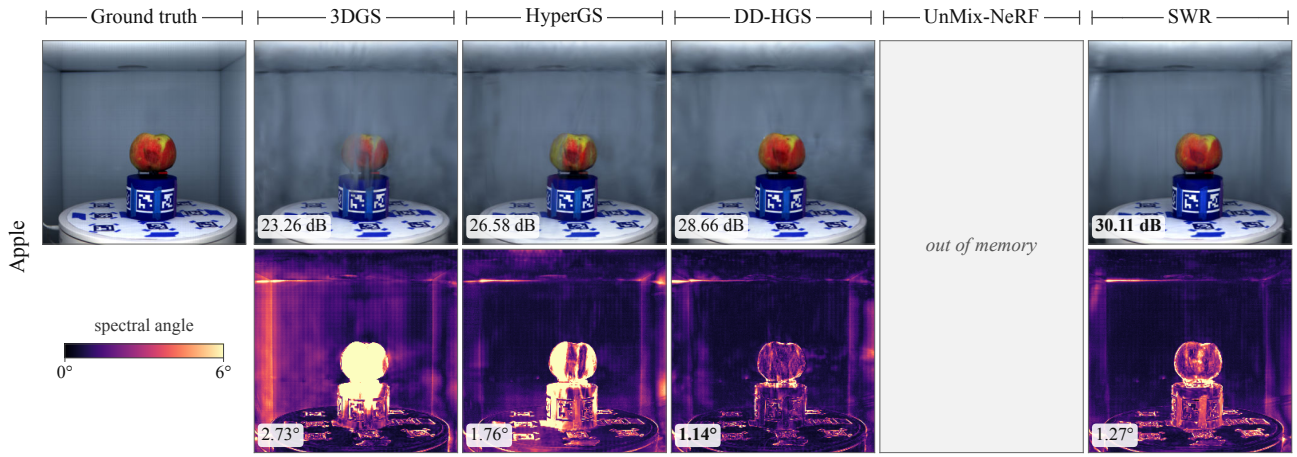


Figure S6. **Qualitative comparison on the HSI-SC apple**, first test view, as in Figure S4. UnMix-NeRF runs out of memory.

Table S9. **Training-time breakdown on City.** Seconds on one RTX 4090, including start-up and loading; seed 42, with SWR averaged over two runs. Rounded SWR stages: anchor fitting (73), RGB feature refinement (37), appearance clustering (68) and spectral decoding (220). Components and totals are rounded independently. UnMix-NeRF jointly fits representation and unmixing.

Method	Representation	Unmixing	Total
UnMix-NeRF	1457		1457
3DGS + VCA/NNLS	5030	71	5102
SWR	110	288	**397**

Query runtime. Spectral comparison and selection rasterization at 960×540 take 3–4 ms on an RTX 3090.

From spectra to a 3D selection. Training-view aggregation gives each Gaussian a spectral descriptor. A query scores these descriptors once, and the resulting scores remain attached to the same primitives as the camera moves. Visibility weights then render the selection from each new view-

Table S10. **Rendering speed** in test views per second on one RTX 4090, at 960×540 with 64 bands, including spectra and label maps. VCA also needs renders of the training views. Half precision skips intersections of weight below 10^{-4}; its dominant labels agree with full precision on 99.97 % of pixels.

	Market	Botanical	Study	Karst	Mean
Baselines					
UnMix-NeRF	0.18	0.21	0.21	0.12	0.18
3DGS + VCA/NNLS	2.53	2.51	4.70	4.73	3.62
including training renders	1.49	1.47	2.18	2.16	1.83
Spectral World Renderer					
Full precision	3.38	3.79	3.52	2.64	3.33
Half precision	8.19	8.81	8.09	6.34	7.86
Half precision, fused kernel	**36.5**	**39.3**	**36.3**	**27.4**	**34.9**

Table S11. **Object-level material retrieval in City.** Mean instance counts distinguish targets lost before spectral matching from false positives introduced by retrieval. Both query types retrieve all available target instances.

Variant	Target trees	Targets lost before retrieval: Material assignment	Targets lost before retrieval: Sparse crowns	Available targets	Candidate trees	False positives↓: SWR	False positives↓: RGB
Mineral leaves	61	14.5	43	3.5	55.5	**0**	6.5
Stressed leaves	87	6.5	62	18.5	61.5	**1**	**1**

A target is unavailable if assigned to another material or represented by too few Gaussians to form a crown instance. False positives are ordinary trees selected incorrectly. Figure 9 reports counts for the illustrated query.

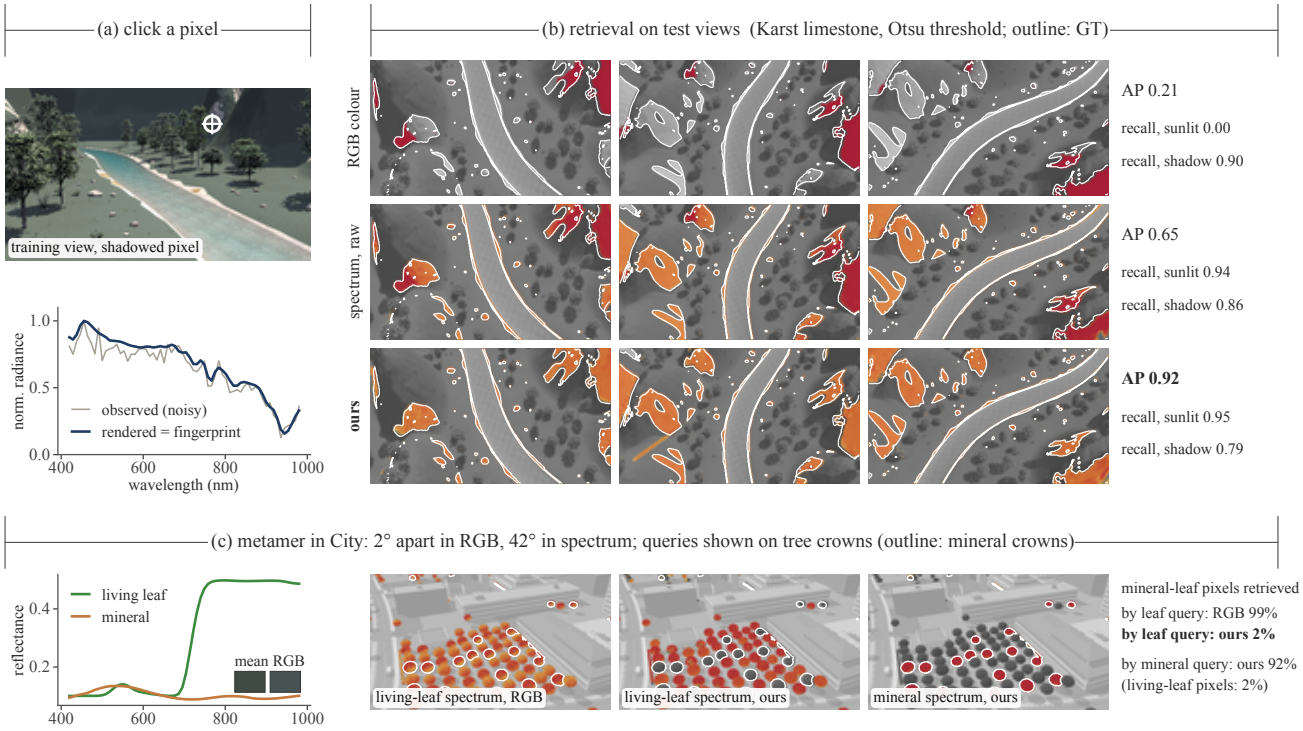


Figure S7. **Material retrieval from clicked and library spectra.** (a) A rendered spectrum clicked on shadowed limestone in a Karst training view defines the query. (b) Test-view selections use Otsu thresholds on RGB, raw spectral and whitened spectral angles; white outlines mark ground truth. AP and recall cover all test views. Whitening retrieves sunlit rock missed by RGB. (c) In the City metamer variant, spectral queries distinguish mineral crowns (outlined) from living foliage; RGB confuses them.

point. This reuses a scene-level spectral comparison across cameras instead of per-image classification.

Pixel and object retrieval. The AP above measures agreement with test-view class masks. Table S11 instead evaluates tree instances after material assignment and crown clustering. These tests address different parts of retrieval: spectral scores distinguish materials, while object coverage also depends on whether enough Gaussians represent a crown to form an instance. The table therefore separates targets lost before spectral matching from false positives of the query.

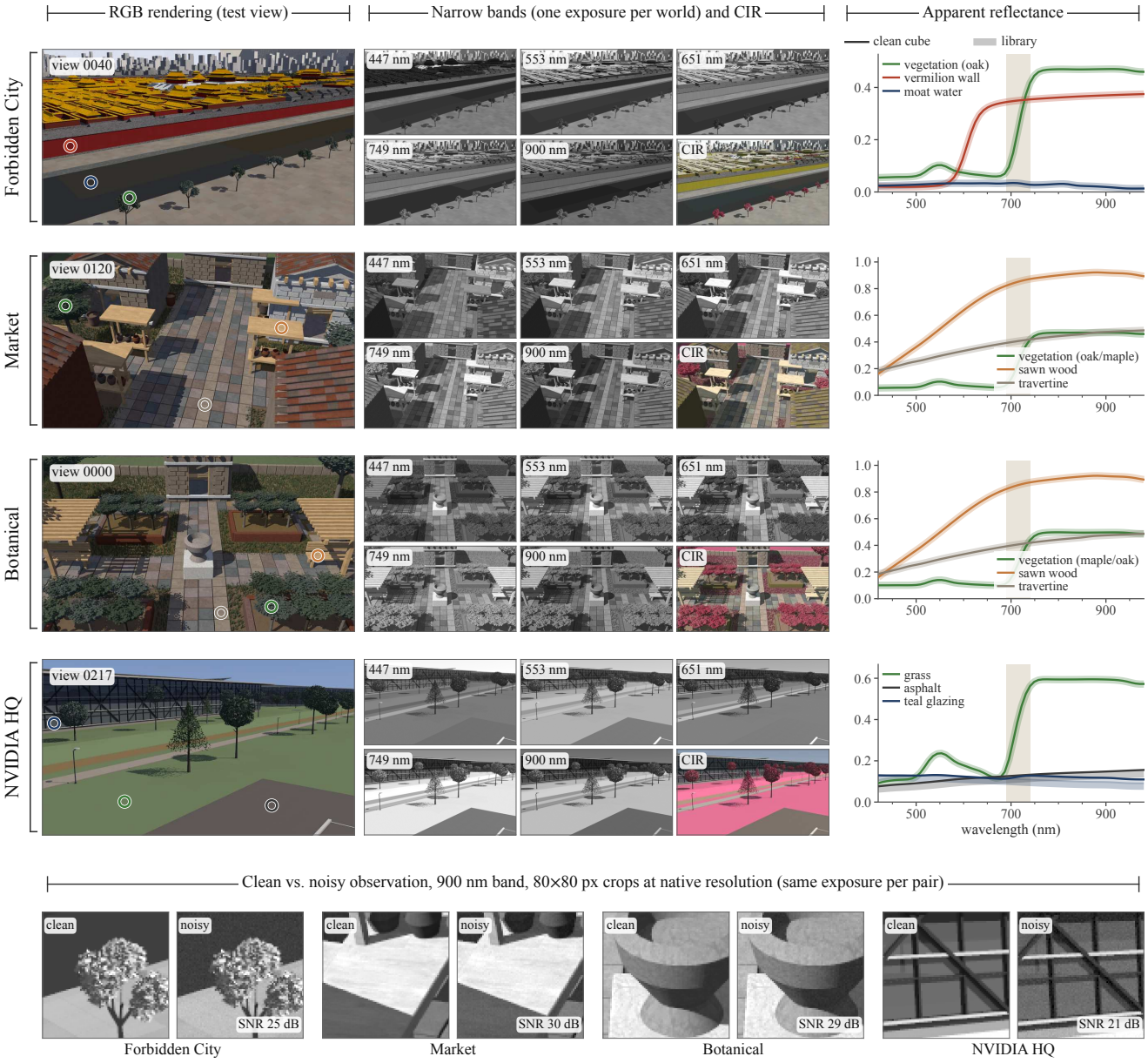


Figure S8. **Simulated hyperspectral observations: Forbidden City, Market, Botanical and NVIDIA HQ.** Narrow bands share one exposure per world, so brightness differences between bands follow the radiance; the colour-infrared composite (R, G, B = 847, 651, 553 nm) renders vegetation red. Apparent reflectance divides the clean radiance by the white-reference radiance under each pixel's illumination (median and 10–90 % range, over the library reflectance of the assigned specimen; shaded: red edge, 690–740 nm). The recovered spectra reproduce the library reflectance, including chlorophyll absorption near 670 nm and the red edge. Bottom: clean and noisy crops of the 900 nm band with the signal-to-noise ratio of the noisy crop.

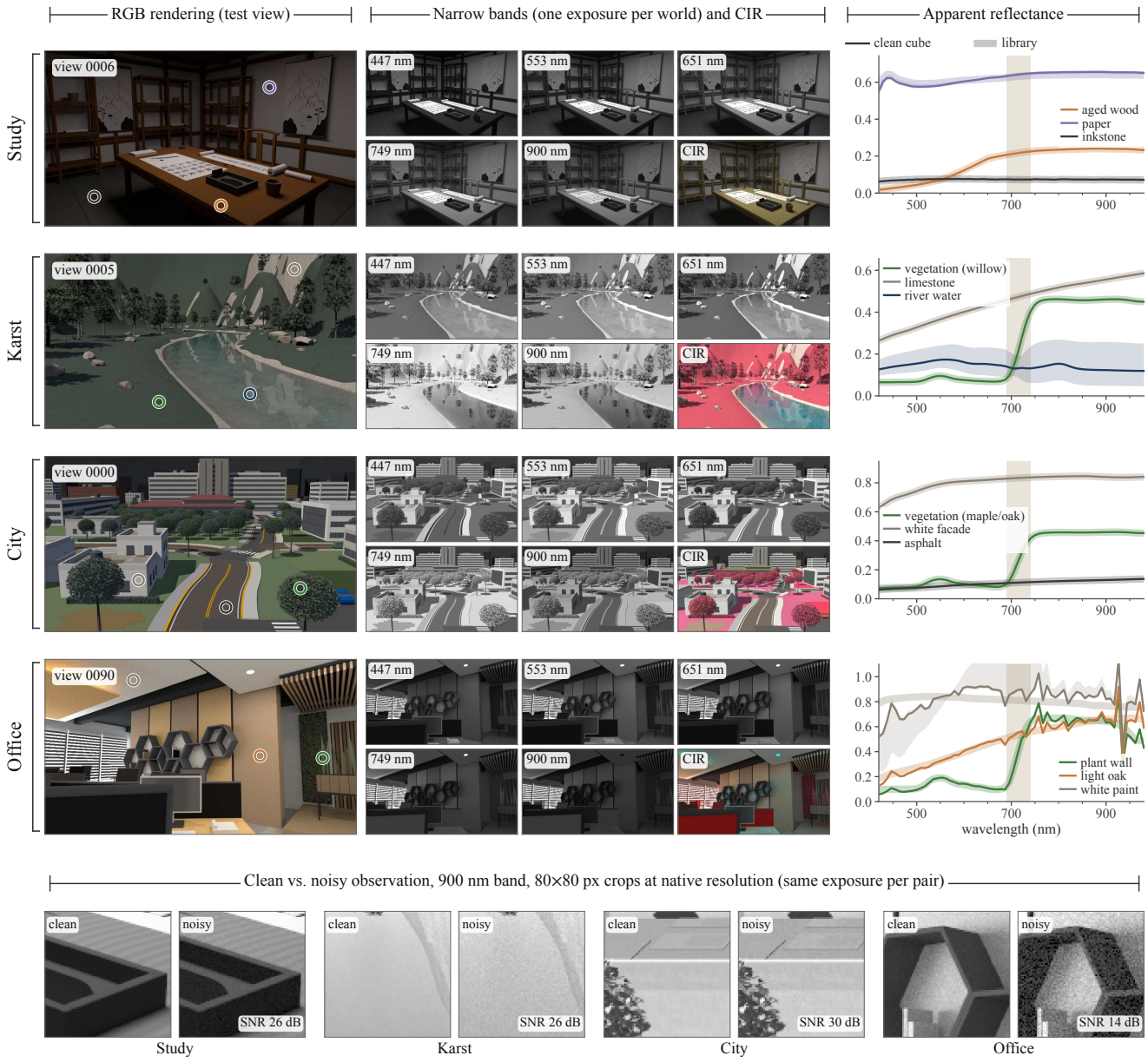


Figure S9. **Simulated hyperspectral observations: Study, Karst, City and Office.** Layout as in Figure S8. Indoor worlds are lit by white sources, so their bands differ less; the Office lights emit no near-infrared, which comes only from daylight through the blinds. Path-traced interiors deviate through interreflection and residual Monte Carlo variance.

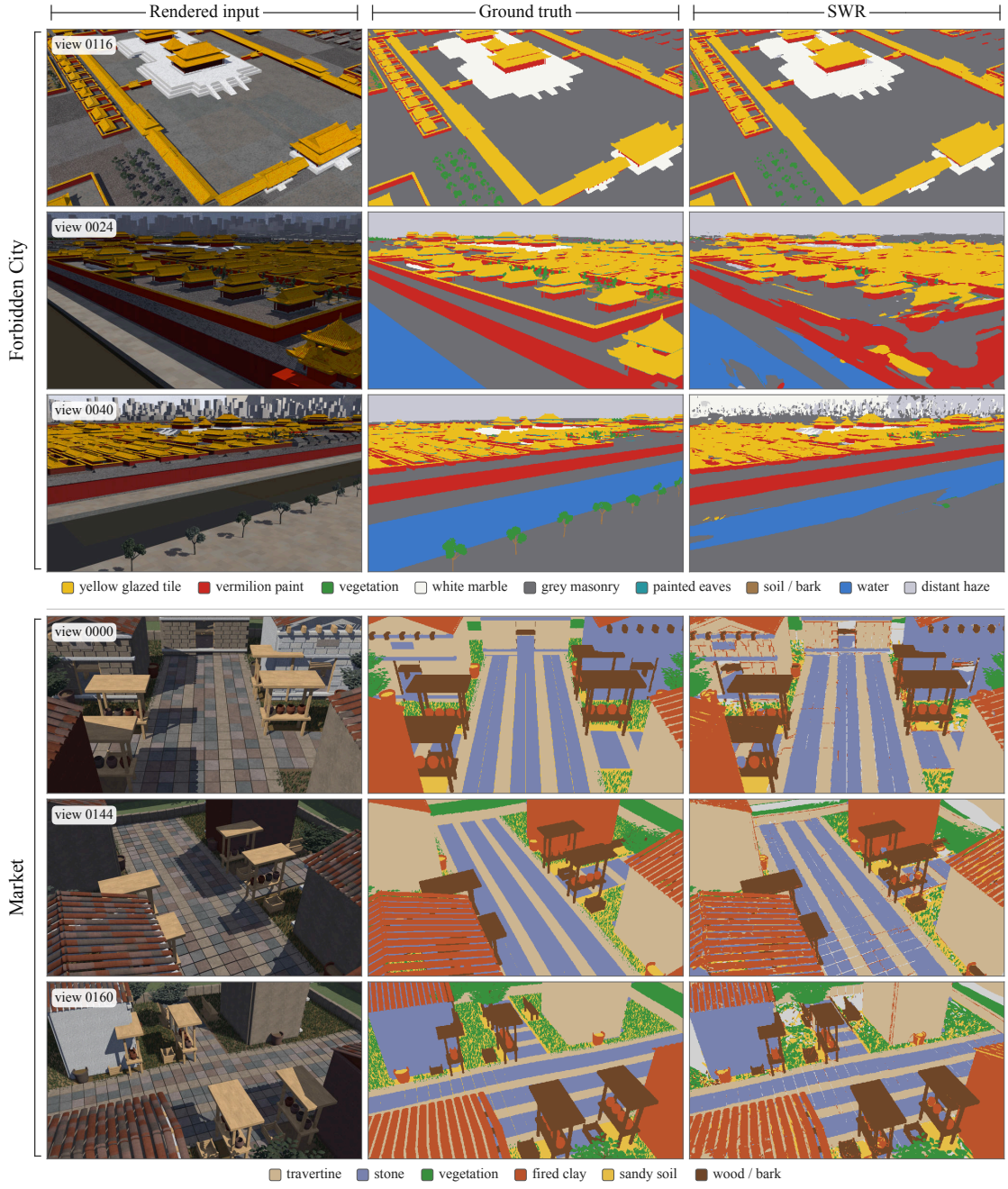


Figure S10. **Additional test views: Forbidden City and Market.** Three separated test cameras per world, shown as full frames.

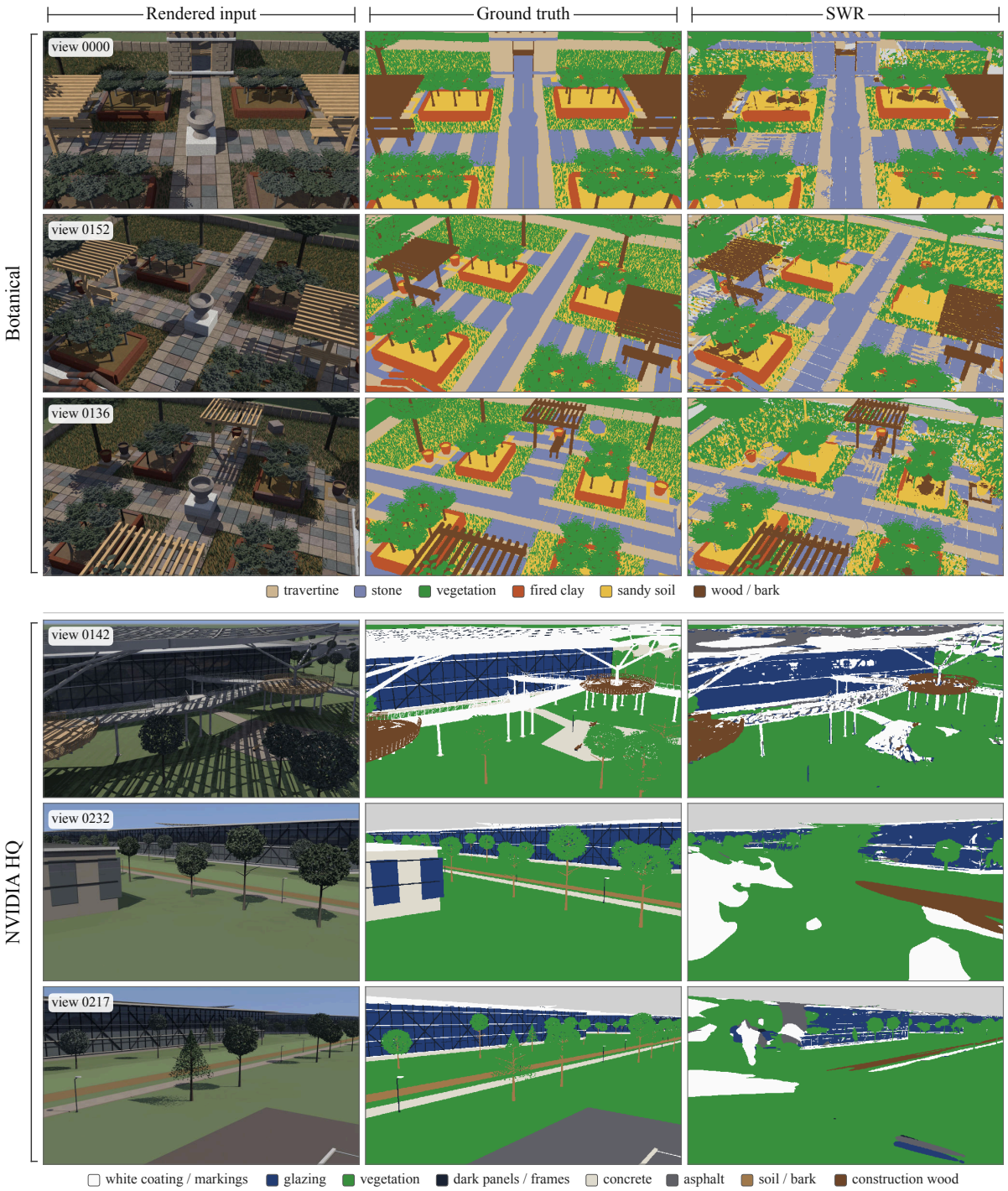


Figure S11. **Additional test views: Botanical and NVIDIA HQ.** Three separated test cameras per world, shown as full frames.

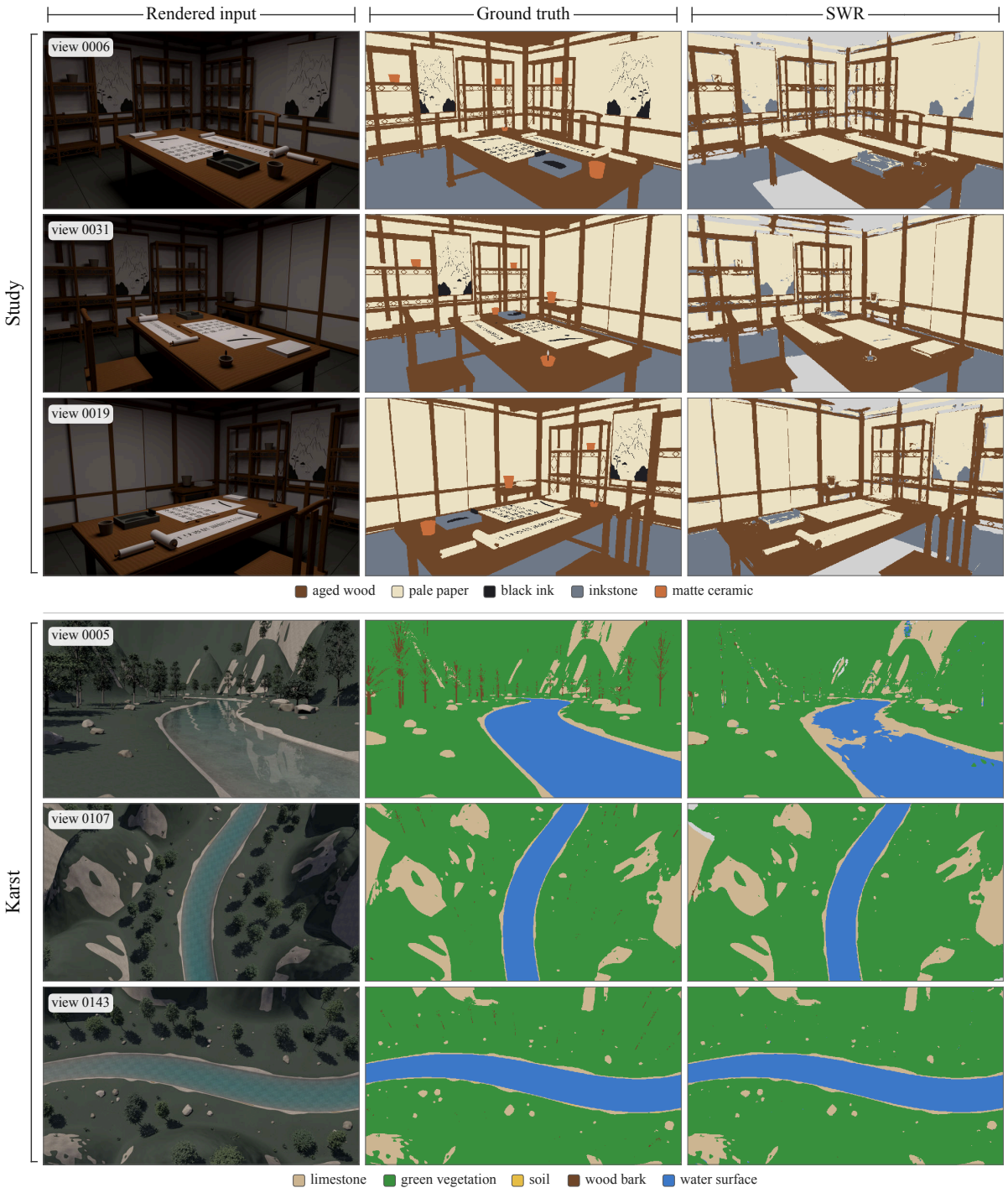


Figure S12. **Additional test views: Study and Karst.** Three separated test cameras per world, shown as full frames.

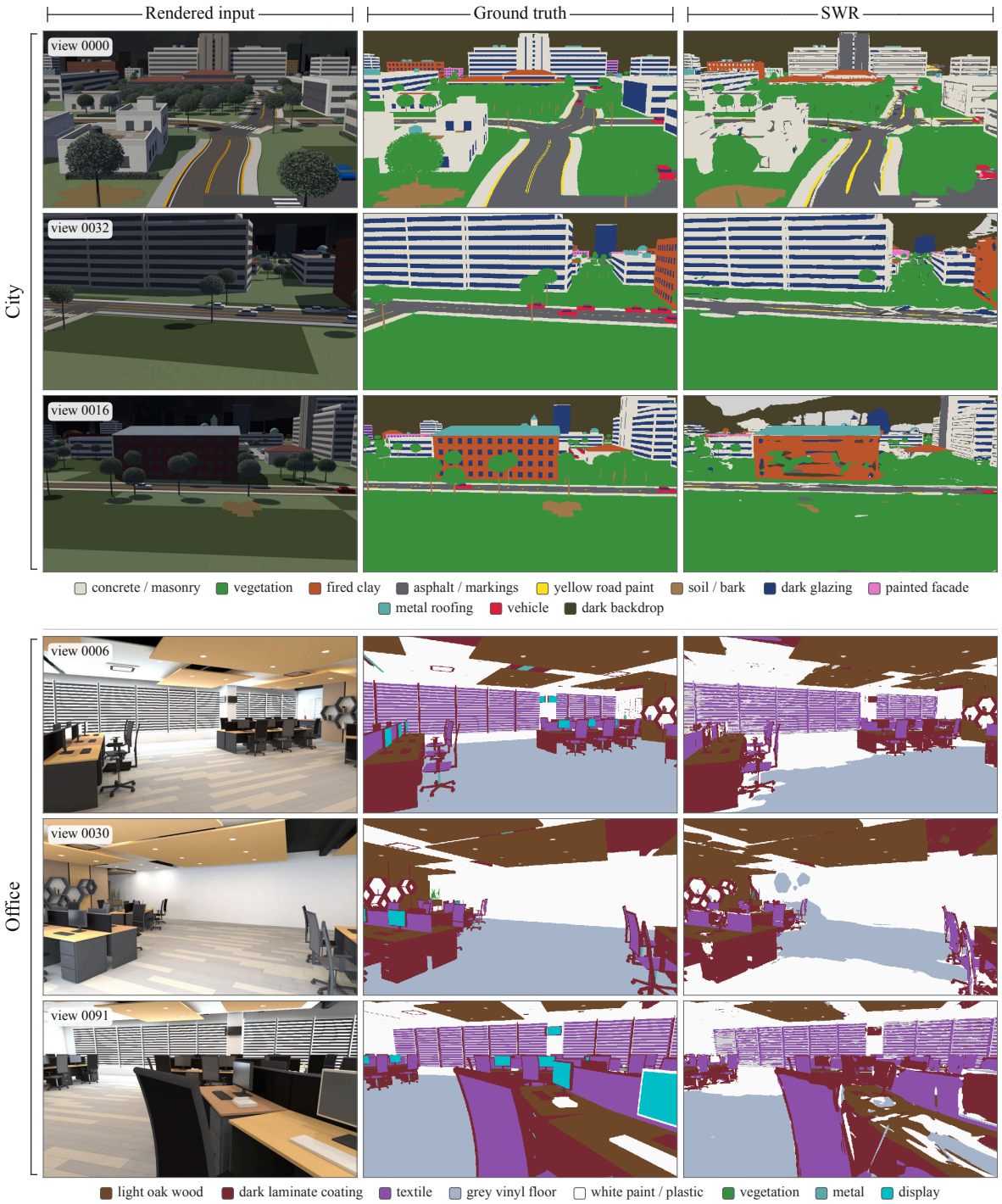


Figure S13. **Additional test views: City and Office.** Three separated test cameras per world, shown as full frames.